Lost Wages and House Prices: The Oil Bust Where Norway's Offshore Workers Lived

Oğuz Yurtoğlu*

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Abstract

What does a loss of wages do to a local economy? This paper uses Norway's offshore oil workers and the 2014 oil price fall to study a shock that reached places through their residents' jobs, where the oil firms were absent. Because these workers spend four weeks at home for every two on the platform, nearly half of them live in municipalities with almost no oil firms. I compare 86 economic regions made up of 375 such municipalities in a difference-in-differences design by their share of employed residents who worked offshore in 2003. From 2013 to 2017, relative to less exposed regions, income from employment fell by 2.2 percent, employment by 1.5 percent, partly continuing an earlier decline, and house prices by 9 percent per percentage point of that share, while the housing stock and the population did not move. Within regions, the income estimates are imprecise but close to the lost offshore pay. Residents' jobs at the oil suppliers fell about as much as their offshore jobs, but the price effect is about as large where few residents worked for the suppliers. Prices fell about four times as much as income. With a fixed housing stock, Norwegian estimates of housing demand imply that a lasting loss of income of the 2017 size, which forecasters expected, accounts for about two-fifths of that fall.

JEL: R31, R23, J23, Q33 **Keywords:** house prices, local labour markets, oil price shock, offshore workers, income shocks, Norway

*Koç University, Department of Economics. Email: oyurtoglu22@ku.edu.tr. Preliminary draft, comments welcome. Replication code is available from the author.

1 Introduction

The 2014 oil price bust has been studied where the oil industry is located, so a region's workers and its firms are hit together. This paper measures it where the workers lived, in municipalities with almost no oil firms, and asks what lost wages do to a local economy. The bust was large. The oil price had risen from $29 a barrel in 2003 to $109 in 2013, and it fell by more than 70 percent between June 2014 and January 2016. In Norway, investment in oil and gas extraction fell by a quarter in current prices from 2013 to 2016 and exploration investment by more than 40 percent. The number of offshore workers whose workplace was the continental shelf fell by a quarter.

Stavanger, home to the headquarters of the largest operator on the continental shelf, Statoil (Equinor since 2018), and areas close to it diverged from the rest of Norway in income, employment and house prices. Between 2013 and 2016, registered unemployment rose by 2.5 percentage points in the county around Stavanger (Rogaland) and barely changed in the rest of Norway. The average sale price of dwellings rose by about 1 percent in Rogaland and by about 21 percent in the rest of Norway.

Research on the bust follows this contrast. Böjeryd (2025) follows homeowners in the Stavanger region, where the bust lowered earnings and house prices. Aastveit et al. (2025) compare government workers in oil and non-oil regions to measure how spending responded to the fall in house prices. Juelsrud and Wold (2025) compare oil engineers with other skilled workers in the oil region, and Schøne and Strøm (2021) compare couples in labour markets with many and few oil jobs.

Work on resource booms and busts elsewhere also measures them where the industry sits. Oil and gas booms raise local wages and incomes (Allcott and Keniston, 2018; Feyrer et al., 2017) and house prices (Bartik et al., 2019). After a bust, incomes fall below the path they would have followed (Jacobsen and Parker, 2016). Each tradable job carries about 1.6 local service jobs with it (Moretti, 2010), and when household wealth falls the jobs that go are the ones that serve local demand (Mian and Sufi, 2014). Notowidigdo (2020) finds that an adverse local shock pushes fewer people out than a favourable one draws in, and estimates how far house prices and transfers carry the loss for those who stay. Commuting spreads a local shock across the places it links (Monte et al., 2018), and studies of fly-in fly-out mining describe the home towns of long-distance workers (Storey, 2001; Haslam McKenzie and Hoath, 2014). The closest study is from Canada, where residents of other provinces work in the oil sands of Alberta. Green et al. (2019) measure the boom of the 2000s where these commuters lived, with a region's commuting rate in 2000 times the oil price, which is close to the exposure I use. Wages rose in the home regions, also for workers who never commuted. Kilian and Zhou (2022) show that oil price shocks moved house prices in Canadian regions without oil, through government transfers and trade between provinces. In most of these places the workers, the firms and the landowners are hit together. I study a bust instead of a boom, in places where the industry had almost no firms, and housing, which Green et al. (2019) and Kilian and Zhou (2022) leave aside as a channel for commuters. The offshore rotation gives a place where the workers are hit and the oil companies are absent. The fall in prices can then be read as the response to lost jobs and pay in the oil value chain, with the exposure of the firm side left in the sample (the equipment

suppliers, the yards and the coastal industries) held fixed.

The places outside the oil regions were not all alike. Few of their local businesses were in the oil industry but some of their residents worked on the continental shelf, in extraction, drilling and the service jobs around them. In 2003, 14,912 Norwegian residents had the continental shelf as their registered workplace. Forty-seven percent of them lived outside Rogaland and Hordaland, the two largest oil counties. They were also spread unevenly.[1] In the 375 municipalities without an onshore oil economy, the share of employed residents working offshore ran from zero in 40 municipalities to more than 3 percent in Tysnes, Fedje, Kvinesdal and Bamble. Over the bust, from 2013 to 2017, the average sale price of dwellings rose by 38 percent in the half of these municipalities with the fewest offshore workers and by 20 percent in the tenth with the most. The average income from employment rose by 10 percent and 6 percent. Compared with Stavanger all of these places did well but compared with one another, those with more offshore workers did worse.

Offshore workers could live far from the industry because their work leaves them long stretches at home. The Working Environment Act of 1977 was applied to the fixed platforms and limited working hours for round-the-clock work. Since then crews have worked two weeks on the platform followed by at least three weeks ashore (Meland, 2019b). After 1986 the pattern became two weeks on, three off, two on and four off. In 2002 and 2003 it became two weeks on and four off, first by local agreements and then under the offshore collective agreements (Meland, 2019a; Knutsen et al., 2025). The agreements also make the employer pay for the trip between the worker's home and the heliport.[2] A worker who spends most of the year on the mainland does not need to live near the heliport. The worker, in nine cases out of ten a man, and his family can live where they choose and many did so long before the boom. In the census of 1980, 773 (35 percent) of the 2,192 residents at work in the North Sea in the census week lived outside Rogaland and Hordaland. The register series starts in 2000. In every year from then to 2017, between 45 and 50 percent of offshore workers lived outside the two counties (Table 1).

The fall in the oil price that hit Stavanger should therefore have reached these regions too, through the earnings of residents who worked at the shelf. Offshore workers lost their jobs wherever they lived, including in towns with no oil firms at all. This paper asks how a shock that reaches a place only through its residents' wages affects its labour and housing markets. The question is not Norway's alone. Wherever people work far from where they live, a shock to a distant employer reaches their home town through their pay, and rotation work makes that exposure measurable. To answer it, I use public municipal data from Statistics Norway and a difference-in-differences design that compares economic regions by how much they were exposed. Exposure is the share of employed residents who worked on the shelf in 2003, the year before the oil price began to rise, and it is held at that value throughout. By 2003 the rotation had long shaped where offshore workers lived. The sample is the 375 municipalities where at most 0.5 percent of employed residents held onshore oil jobs at the oil companies and their service firms in 2003. I count the jobs in the municipality and

[1] Table 9 lists the ten municipalities with the most offshore work and the ten with the least.

[2] The agreement for the mobile rigs has done so since 1974 (Arbeidsretten, 2017), and the operators' agreement of 2024 states it in section 15.1 (Offshore Norge, 2024).

the jobs their residents commuted to elsewhere, so the bust could hardly reach them through oil firms, local or nearby. They hold almost half of the offshore workers and make up 86 economic regions. Most of the variation in offshore work lies between these regions, and the labour and housing markets that the bust worked through reach beyond a municipality, so each region is one observation. A second comparison sets municipalities against their neighbours in the same region. The design assumes that, had the oil price not fallen, regions with more offshore workers in 2003 would have followed the same path as the others.

Table 1: Residents whose workplace is the continental shelf, 1980 and 2000–2017

Year	Norway	Rogaland	Hordaland	Elsewhere	Elsewhere (%)
1980 (census week)	2,192	929	490	773	35.3
2000	15,477	4,969	3,605	6,903	44.6
2003	14,912	4,294	3,658	6,960	46.7
2005	16,290	4,842	3,842	7,606	46.7
2010	17,454	5,059	4,154	8,241	47.2
2013	20,603	5,609	4,840	10,154	49.3
2015	17,842	4,767	4,261	8,814	49.4
2017	14,742	4,035	3,624	7,083	48.0

Note: 1980: residents aged 16 and over who were at work in the week of 25–31 October 1980 and whose meeting place was the North Sea, from the full count of the 1980 census (table 9 of Statistics Norway's municipal census booklets). Workers ashore on their rotation that week are not counted, so the 1980 row is comparable with the later rows in its shares, not its levels. A booklet lists the North Sea only when it is among the municipality's main destinations. It is listed by 23 of the 26 municipalities of Rogaland, 19 of the 34 of Hordaland and 69 of the 394 elsewhere, so the municipalities left out lie mostly outside the two counties; the row sums the 110 whose numbers can be read. 2000–2017: employed residents whose workplace is the continental shelf in the fourth quarter, by county of residence on the 2019 map (Statistics Norway table 03321). The register series starts in 2000, and later years are not shown because Statistics Norway stopped recoding offshore establishments to the shelf after 2018. The 1990 census reports the North Sea only from a sample in municipalities with 6,000 or more residents and suppresses its small cells, so it is not shown.

From 2013 to 2017, the average income from employment fell by 2.2 percent relative to other regions, employment of residents by 1.5 percent, partly continuing an earlier decline, and house prices by 9 percent per percentage point of employed residents working offshore in 2003. House prices are measured by the average sale price of dwellings. The price per square metre of detached houses fell about as much, although it was not flat before the bust. The dwelling stock and the population did not move during the oil price fall. Prices fell about four times as much as income. The income and price effects hold with the coastal industries and the distance to a heliport held fixed, and the price effect holds when the trend of house prices from 1992 to 2002 is allowed to continue. In the regions with no onshore oil job at all, income and the price per square metre fall by similar amounts, less precisely.[3] Inside a region, municipalities with more offshore workers lost income close to the pay of the offshore jobs they lost, although these estimates are imprecise, and their house prices did not fall detectably more than their neighbours'. The loss of income was the men's. Women's income did not fall during the period. For each offshore job lost, residents lost

[3] Tables 4 and 7.

about one job at the equipment suppliers of the industry, since the sample rule does not remove them. Perhaps half of these jobs were in the region itself, so the shock is to residents' jobs in the oil value chain, most of it offshore pay. The price effect is about as large in the regions with the fewest supplier workers. The rest of the local economy lost about two jobs for each offshore job, although that estimate cannot be told from zero. Whether laid-off offshore workers found other jobs cannot be seen in municipal data, but employment outside oil fell rather than rose, so other industries did not add jobs on net. With the stock of dwellings fixed, Norwegian estimates of housing demand imply that a lasting loss of income of the 2017 size, which forecasts made in 2015 and 2016 expected, accounts for about two-fifths of the fall in prices. Prices kept falling after income began to recover, so much of the rest looks like slow adjustment.

Measuring the shock where the workers lived turns a shock to the oil industry into a shock to residents' jobs and pay, in places with almost no oil firms. Across regions it moves house prices. The equipment suppliers also lost jobs, but the price effect does not depend on how many residents worked for them.

2 Setting

Much of the oil wealth belongs to the state and goes into a sovereign fund. The fund held about 865 billion dollars at the end of 2014.[4] A fiscal rule lets the government spend about the expected real return of the fund each year, so a fall in the oil price lowers what flows into the fund and not what the budget can spend. The government leaned against the downturn with the budgets of 2015 and 2016, Norges Bank cut the policy rate from 1.5 percent to 0.5 percent between December 2014 and March 2016, and the krone lost about a fifth of its value against the import-weighted index (International Monetary Fund, 2016). These responses reached every region, and the comparison across regions removes their common part. Their effects could still differ between regions. The weaker krone helped the coastal exporters, and the robustness checks hold their industries fixed, along with the national cycle of house prices (Section 6.3).

2.1 Offshore work

Norway's oil and gas lie under the continental shelf, and the onshore oil economy grew up around Stavanger and Bergen (Figure 1). In 2003 the extraction of oil and gas and the services around it employed 28,775 people, 16,593 of them on land.[5] The Stavanger/Sandnes region held 59 percent of the onshore jobs. The other 12,182 jobs were on the shelf itself, which Statistics Norway records as a place of work of its own. With the platform work and catering registered there as well, 14,912 residents had the shelf as their workplace. Crews flew from five heliports, so the workplace fixes the heliport a worker flies from but not where the worker lives. Where they live is not random. Within

[4]The fund held 6,431 billion kroner. Dollars are converted at Norges Bank's rate on 31 December 2014 of 7.43 kroner per dollar.

[5]Division 11 of the industry classification SN2002.

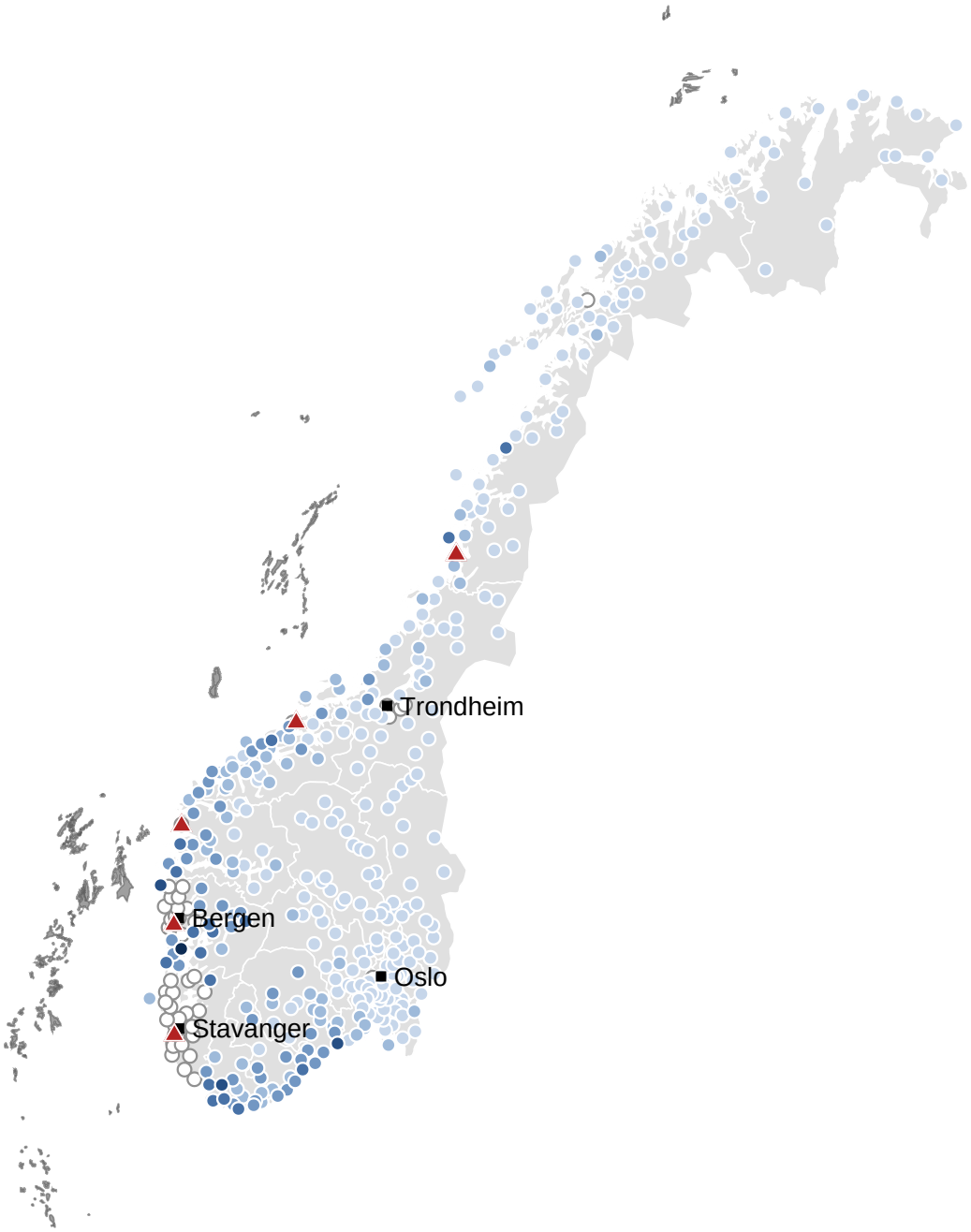


Figure 1: Oil and gas fields, offshore work and the sample

Note: The grey areas at sea are the outlines of the 139 oil and gas fields in the Offshore Directorate's FactMaps in October 2026, which include fields developed after 2003. Triangles mark the five heliports from which crews flew offshore in 2003: Stavanger (Sola), Bergen (Flesland), Florø, Kristiansund and Brønnøysund. Each point on land is a municipality, at its administrative centre. Points in the sample are shaded by the share of employed residents whose workplace is the continental shelf in the fourth quarter of 2003: from lightest to darkest below 0.5 percent, 0.5 to 1, 1 to 2, 2 to 3, 3 to 4, and 4 percent or more. White points are the 46 municipalities outside the sample, where more than 0.5 percent of employed residents held onshore oil jobs, in the municipality or by commuting (Section 3.2). Lines inside Norway are the 18 counties of 2019. Squares mark four towns. Tønsberg and Sandefjord, which are one unit in the data because Stokke was split between them in 2017, are both drawn.

the sample and weighted by population, the 2003 share correlates −0.53 with the log distance to the nearest heliport and 0.44 with the share working in ship and platform building, and the robustness checks hold these coastal traits fixed.

2.2 Boom and bust

The oil price averaged between 25 and 29 dollars a barrel from 2000 to 2003 and stayed around 110 from 2011 to 2013 (Figure 2). Investment on the shelf rose from 13 billion dollars in 2003 to 37 billion in 2013, at 2013 prices.[6] The number of residents working on the shelf rose from 14,912 in 2003 to 20,603 in 2013. I measure the exposure in 2003, before the boom moved it. The bust began in June 2014 as rising oil production met weaker global demand (Baumeister and Kilian, 2016). The number of residents working on the shelf fell from 20,603 in 2013 to 14,742 in 2017. The cuts fell on exploration, drilling, engineering and supply. These are the activities that employ rotation crews. The loss of offshore pay therefore arrived from 2014 in every municipality that housed offshore workers.



(a) Oil price

(b) Planned and invested petroleum investment

(c) Residents working on the shelf

Figure 2: The boom and the bust, 2000–2019

Note: Panel (a) is the annual mean of the daily Brent spot price in US dollars per barrel (US Energy Information Administration, through FRED). Panel (b) shows accrued investment in oil and gas extraction and pipeline transport (solid line, filled markers; Statistics Norway) and the investment that the oil companies had planned for the same year in the May survey of the year before (dashed line, hollow markers). Both are in billion dollars of 2013, deflated with the consumer price index and converted at Norges Bank's average rate for 2013 of 5.88 kroner per dollar. Panel (c) is the number of residents whose workplace is the continental shelf in the fourth quarter, shown to 2017 (Section 3.1). The dashed lines fall between 2003 and 2004 and between 2013 and 2014.

2.3 Layoffs and benefits

A worker who lost an offshore job received the same unemployment insurance as any other worker, 62.4 percent of previous earnings up to a cap of six times the base amount, for up to two years. Employers could also lay workers off temporarily, and the employment register counts a worker on

[6] From 77 to 219 billion kroner at 2013 prices, deflated with the consumer price index. Dollars are converted at Norges Bank's average rate for 2013 of 5.88 kroner per dollar.

temporary layoff as employed if the layoff is expected to last less than 90 days. The income measure leaves benefits out, so it records the loss of pay before insurance.

3 Data

All data used in this paper are retrieved from public sources of Statistics Norway's StatBank, the Labour and Welfare Administration (NAV) and the Norwegian Offshore Directorate.[7] Municipality level data is used to build the regions that the regressions compare, and each year's data is carried forward to 2019 with Statistics Norway's official list of code changes. If municipalities merged during the sample period, the merged unit carries the whole population throughout the sample. The only split between 2000 and 2019 is that of Stokke in 2017, so I merge its two successors (Tønsberg and Sandefjord) into one unit. This leaves 421 municipalities in 88 economic regions.

3.1 Exposure

Statistics Norway records the continental shelf as a place of work of its own and links each worker to the municipality where they live. For each municipality I divide the number of residents whose workplace is the shelf by the number of employed residents. In 2003, the workplace of 14,912 Norwegian residents was registered as the shelf. Of these, 82 percent worked in the extraction of oil and gas and the services around it[8], 10 percent in platform and yard work and 6 percent in catering. I fix the exposure in 2003, the last year before the oil price began to rise, so exposure is the share of a municipality's employed residents whose workplace was the continental shelf in the fourth quarter of 2003.

The count of offshore workers is a lower bound. Employers register their establishments at fixed addresses, and Statistics Norway moved a job to the shelf only when it knew the establishment worked offshore. These jobs are mainly in oil services, drilling and catering. Offshore workers of employers registered at an onshore office stay at that office's address, workers on mobile rigs are recorded as seafarers. Foreign residents and commuters from Sweden have no home municipality at all. If every municipality's count misses the same fraction of its offshore workers, they overstate the effect of one true point by that fraction. If the fraction varies across municipalities for reasons unrelated to the outcomes, it weakens them. The share calculated from the 2000 data correlates 0.93 with its 2003 value. The 1980 census counts only the workers who were offshore in the census week and lists the North Sea for a hundred municipalities. With the unlisted municipalities at zero, its share correlates 0.7 with the 2003 share on the sample and 0.8 across all municipalities.[9] Therefore, the shares are assumed to be stable before the boom. The series changes source in 2015 and breaks again in 2018, both well after the year I use to calculate the exposure.

[7] Table 20 in the appendix lists every table.

[8] Division 11 of the industry classification SN2002.

[9] Residents at work in the North Sea in the census week of 1980 per employed resident of 2003, on the 2019 map, weighted by population. The census count is described in the note to Table 1.

In 2003, the share of employed residents working offshore was zero in 40 municipalities and 0.28 percent at the median (Table 3). It was 1.33 percent at the 90th percentile and above 3 percent in four municipalities (Table 9). Weighted by population, the mean is 0.41 percent and the standard deviation 0.57. The coefficients in this paper are per percentage point of exposure, which is about the difference between the 90th and the 10th percentile. Across the 86 economic regions of the sample, the regional share runs from 0.08 percent at the 10th percentile to 1.51 percent at the 90th. Figure 1 maps the share across Norway.

3.2 Sample

The sample is restricted to municipalities whose residents held almost no onshore oil jobs in 2003 since the design needs places that the bust could reach mainly through their residents. For each municipality I count its employed residents by the municipality where they work and weight each of them by the share of oil extraction and related jobs among all jobs located in that workplace municipality. The sum includes the residents who work in their own municipality, so the measure covers oil firms in the municipality as well as commuting to oil jobs elsewhere. Divided by the municipality's employed residents, it estimates the share of its workers who held onshore oil jobs, on the assumption that a commuter holds an oil job as often as the average worker where they work. Jobs on the continental shelf are left out, since they are the exposure. Across all municipalities the measure allocates the 16,593 onshore oil jobs to the places where their holders lived. The cutoff is set by a gap in the data. In 344 of the 421 municipalities the share was at most 0.2 percent, and in the median one it was 0.07 percent. Above 0.2 percent the values rise without a break to 0.48 percent in Porsgrunn. The next one is Etne with 0.57 percent. I keep the municipalities at or below 0.5 percent. The larger gaps above 1.2 percent would keep the oil-office cities and their commuter suburbs, such as Bærum, Trondheim, Askøy and Karmøy. The rule leaves out the oil centres of Stavanger, Haugesund and Bergen with their commuter belts which corresponds to 46 municipalities. The sample is the remaining 375 municipalities in 86 economic regions. They had 3.8 million inhabitants in 2014, three quarters of the population. They also held 7,134 (48 percent) of the 14,912 offshore workers.

A rule based on the oil jobs located in the municipality would not do the same job. It would keep 30 municipalities that this rule leaves out. Most of them are suburbs and islands around Stavanger, Haugesund and Bergen, such as Gjesdal, Rennesøy, Strand, Klepp and Time. They have few oil offices of their own, but their residents commute to the offices next door. In these 30 municipalities, 1.5 percent of employed residents held onshore oil jobs and 2.3 percent worked offshore. In the sample the two shares are 0.1 and 0.41 percent. Including them would put the places most exposed to the onshore oil economy at the top of the offshore distribution.

Division 11 is a narrow definition of the oil jobs. It leaves out the yards, the equipment makers, the supply ships and the supply bases. These were also hit in 2015. I therefore build a second rule for the supply chain, measured in the same year as the exposure. Statistics Norway's list of petroleum-related industries exists only from 2008, five years into the boom. In that year 90 percent

of its jobs fell into two divisions of the older classification. These are division 11 and division 35, which covers ships and platforms.[10] Both divisions are published by place of work for 2003 and their share of a region's jobs in 2003 correlates 0.98 with the share on the 2008 list. A second check uses the 2008 list directly. A municipality is a hub if at least 5 percent of the jobs located in it are on the list, or at least 3 percent in a stricter version. Most hubs lie on the coast between Sunnhordland and Trøndelag. Because workers commute within economic regions, the check leaves out every municipality in a region that contains a hub. That leaves 272 of the 375 municipalities at 5 percent and 240 at 3 percent, with 58 and 47 percent of the sample's offshore workers. Table 2 shows this rule and the other samples I use to test the choice of sample, and Section 6 reports the estimates on each.

Table 2: The headline sample and the alternative samples

			Population	Offshore workers		Offshore work (%)	
Sample	Munic.	Regions	(1,000)	Number	% of all	Mean	P90
All municipalities	421	88	5,109	14,912	100	0.67	2.04
Headline sample	375	86	3,849	7,134	48	0.41	1.12
Onshore jobs located in the region only	360	79	3,778	7,358	49	0.44	1.18
Outside Rogaland and Hordaland	362	80	4,144	6,960	47	0.38	0.89
No onshore oil job in the region	259	59	1,721	3,138	21	0.40	0.98
Regional onshore share $\leq 1.2\%$	381	84	4,285	8,211	55	0.43	1.18
Municipal onshore share $\leq 0.5\%$	404	88	4,089	9,513	64	0.53	1.44
Supply chain 2003 $\leq 1\%$	256	56	3,257	3,942	26	0.27	0.77
Supply chain 2003 $\leq 2\%$	296	66	3,506	4,998	34	0.32	0.78

Note: Population on 1 January 2014. Offshore workers are residents whose workplace was the continental shelf in the fourth quarter of 2003, and offshore work is their share of employed residents; the mean and the 90th percentile are weighted by population. The headline rule counts the onshore oil jobs held by a municipality's residents and the regional rule the onshore oil jobs located in its region. The supply-chain rule counts the region's jobs in SN2002 divisions 11 and 35. All three are measured in 2003. The rows below the headline sample apply each rule to all 421 municipalities. Inside the headline sample, the regions with no onshore oil job hold 256 municipalities in 58 regions, and the supply-chain rules keep 250 and 290 municipalities in 55 and 65 regions.

3.3 Outcomes

The outcomes follow the shock from income to the labour market and to housing (Table 3). The first stage is the average income from employment per resident aged 17 and over, from the tax records from 2005 onwards. It leaves benefits out, so it measures the loss of pay before insurance and taxes, and because it is measured per resident it moves with both the pay of those in work and the share of adults who work. When municipalities merged, I average their means with population weights. To see whose income fell, I also use it for men and for women separately, since nine in ten of the offshore jobs of 2003 were held by men. As a check I use the median household income after tax, which includes unemployment benefits and so measures the loss after insurance and taxes, averaged with household weights when municipalities merged.

[10] Statistics Norway's correspondence table between SN2007 and SN2002.

For the labour market I use the number of employed residents aged 20 to 66 in the fourth quarter and the registered unemployment rate in November. The employment register moved to a new source in 2015. Statistics Norway gives no overlap year or back-cast by residence, but it published the 2015 count of employees by workplace municipality twice, on the new source and adjusted to the old one. The gap between the two is the size of the break in each municipality. It does not follow offshore work. I therefore measure employment from 2013 to 2017, like income and prices.[11]

To see which jobs were lost, I also use employed residents by industry of their workplace from 2013 to 2019.

For housing I use the average price per sale of dwellings sold on the free market, the dwelling stock on 1 January and the number of dwellings given a building permit. The average price is total purchase value divided by the number of sales, so I use it only where a municipality had at least 10 sales in every year from 2003 to 2019, which keeps the panel balanced. That leaves 271 of the 375 municipalities, with 96 percent of the sample's population and 95 percent of its offshore workers. The average price is not adjusted for the size or type of the dwellings sold, so I report the price per square metre of detached houses next to it. Statistics Norway publishes it only where there were enough sales. It is observed in 2013 and 2017 in 225 of the 375 municipalities, in 82 of the 86 regions. Population is the count on 1 January.

The bust windows start in 2013, the last year before the price fell. Income, employment and prices are measured from 2013 to 2017 and registered unemployment from November 2013 to November 2016. The dwelling stock and the population are measured from 1 January 2014 to 1 January 2018. The windows end in 2017 because investment on the shelf and the number of offshore workers reached their lowest level that year and the oil price had begun to recover. The series that counts offshore workers also breaks in 2018, when Statistics Norway stopped recoding offshore establishments to the shelf. The event studies run from 2008 to 2019. For the boom the windows run from 2003 to 2013, or from 2005 for income and from 2006 for the dwelling stock, where those series begin. Every outcome is 100 times the change in logs over its window. Registered unemployment is the change in percentage points.

3.4 Controls

Every regression holds fixed the size of the municipality and its industry mix before the boom. These are the log population at the start of the window and the 2003 shares of employed residents in industry and mining, in the primary industries and in trade, hotels and restaurants. It also holds fixed two measures of the onshore oil economy and the equipment suppliers. The first is the onshore oil share of the jobs in the municipality's region, since some sample municipalities lie in oil regions. The second is the share of its own residents who held onshore oil jobs, the measure behind the sample rule. The suppliers are measured by the 2008 share of residents working in fabricated metal products and in machinery and equipment, the industries that supplied the oil industry, since the sample rule removes the oil companies and their service firms but not their suppliers. In

[11] Table 15 reports the test.

Table 3: Summary statistics, headline sample

			Headline sample				All
Variable	Unit	*N*	Mean	P10	Median	P90	Mean
Exposure and sample rules, 2003							
Offshore work	% employed residents	375	0.41	0.00	0.28	1.33	0.67
Onshore oil economy of the region	% of the region's jobs	375	0.07	0.00	0.00	0.03	0.75
Residents' onshore oil jobs	% employed residents	375	0.10	0.02	0.06	0.19	0.77
Supply chain of the region	% of the region's jobs	375	0.98	0.01	0.35	3.62	2.09
Controls							
Fishing and fish farming	% employed residents	375	0.83	0.00	0.19	8.86	0.72
Ship and platform building	% employed residents	375	1.02	0.05	0.42	3.29	1.37
Water transport	% employed residents	375	0.88	0.08	0.56	3.41	0.97
Equipment suppliers (25, 28), 2008	% employed residents	375	1.76	0.13	1.28	4.09	1.73
Distance to the nearest heliport	km	375	261.82	58.29	192.07	445.19	215.76
Population, 1 January 2014	thousands	375	10.26	1.15	4.33	20.60	12.14
Outcomes, bust							
Mean employment income per resident, 2013–17	log points	375	8.76	4.24	9.11	12.89	7.76
Median household income after tax, 2013–17	log points	375	9.71	5.80	9.67	13.43	9.00
Employed residents, 2013–17	log points	375	1.22	−7.55	−1.93	3.09	0.79
Registered unemployment, Nov 2013–16	pp	373	−0.11	−1.20	−0.10	1.08	0.26
Average price per sale, 2013–17	log points	271	27.68	8.24	26.27	46.41	24.15
Detached-house price per m^2, 2013–17	log points	225	22.48	5.36	20.45	31.80	18.92
Dwelling stock growth, 2014–18	log points	375	4.66	0.59	3.57	7.38	4.96
Population growth, 2014–18	log points	375	3.42	−3.53	0.74	5.40	3.55
Economic regions of the headline sample							
Offshore work	% employed residents	86	0.41	0.08	0.31	1.51	
Equipment suppliers (25, 28), 2008	% employed residents	86	1.76	0.38	1.70	3.76	
Mean employment income per resident, 2013–17	log points	86	8.76	4.45	8.95	11.63	
Employed residents, 2013–17	log points	86	1.22	−4.49	−1.40	2.08	
Average price per sale, 2013–17	log points	86	27.60	13.40	25.95	42.22	
Detached-house price per m^2, 2013–17	log points	82	22.02	6.45	18.46	29.43	
Population, 1 January 2014	thousands	86	44.76	7.94	23.22	92.26	

Note: Municipalities in the headline sample (375) and in Norway (421). Means are weighted by population on 1 January 2014, except for population itself; the percentiles are across municipalities. Log points are 100 times the change in logs and pp are percentage points. The average price per sale is shown for the 271 municipalities of the balanced panel, and the detached-house price where Statistics Norway publishes one. The supply chain of the region is jobs located in the region in SN2002 divisions 11 and 35, as a share of all jobs there, in 2003. The last panel describes the 86 economic regions of the headline sample, each the population-weighted mean of its sample municipalities; means are weighted by regional population and percentiles are across regions.

the regressions on regions each control is the population-weighted mean over the region's sample municipalities. Offshore workers live near the heliports and in communities that already worked at sea, so a second set of controls measures those traits in 2003. These are the shares of employed residents in fishing and fish farming, in ship and platform building and in water transport, and the log distance from the municipality's administrative centre to the nearest of the five heliports in use in 2003.

A third control, used in the robustness checks, is the distance to the nearest of Oslo, Bergen, Trondheim and Stavanger.

4 Empirical strategy

Most of the variation in offshore work lies between economic regions. Weighted by population, 81 percent of the variance of the 2003 share in the sample lies between the 86 regions and 19 percent within them. The bust also worked through markets that are larger than a municipality. A household that lost offshore pay looked for work, spent its income and sold its house in a market that reaches beyond its own municipality, and commuting spreads a local shock across the places linked by it (Monte et al., 2018). I therefore compare regions. Each economic region is one observation, built from the sample municipalities it contains, so the municipalities that the sample rule leaves out do not enter it. A regression on municipalities with standard errors clustered by region makes the same comparison without saying so.

$$\Delta\bar{y}_r = \alpha + \beta\,\bar{S}_r + X_r'\gamma + u_r \tag{1}$$

$\Delta\bar{y}_r$ is the change in the outcome in region r over its bust window, the mean of the changes in the region's sample municipalities weighted by their population on 1 January 2014. Counts, such as jobs, sales and benefit recipients, are summed over the municipalities first. Income, employment and prices are measured from 2013 to 2017 and the dwelling stock and the population from 1 January 2014 to 1 January 2018. Every change is 100 times the change in logs. $\bar{S}_r$ is the region's offshore work, the share of employed residents whose workplace was the continental shelf in the fourth quarter of 2003, in percentage points, averaged over the region's sample municipalities with population weights of 1 January 2004. X_r holds seven controls measured before the bust. Two describe the onshore oil economy. They are the onshore oil share of the jobs in the region and the regional mean of the share of residents who held onshore oil jobs, the measure behind the sample rule. Four describe the region's size and industry mix. They are the log population on 1 January 2014 and the regional means of the 2003 shares of employed residents in industry and mining outside oil, in the primary industries and in trade, hotels and restaurants. The last is the regional mean of the 2008 share of residents working in fabricated metal products and in machinery and equipment, the equipment suppliers of the industry, which the sample rule does not remove. α is a constant and u_r the error. The regressions are weighted by regional population on 1 January 2014, so β is the effect for the average resident per percentage point of offshore work.

4.1 Event studies

The event study makes the same comparison year by year.

$$\bar{y}_{rt} = \lambda_t + \sum_{k\neq 2013} \mathbf{1}[t=k]\,(\beta_k\,\bar{S}_r + X_r'\gamma_k) + u_{rt}, \qquad \bar{y}_{r,2013} = 0 \tag{2}$$

The outcome $\bar{y}_{rt}$ is the change in region r from 2013 to year t, so it is zero in 2013. λ_t is a year effect, and each control has its own coefficient γ_k in every year. The coefficient β_k is the cumulative effect of the bust up to year k after 2013. Before 2013 it tests whether exposed regions were already

diverging from the others. The regressions are weighted by regional population on 1 January 2004 and use the log population of that date. The panel covers 2005 to 2019 for income, 2008 to 2019 for employment and 2003 to 2019 for the average sale price.

4.2 Within regions

The second comparison holds the region fixed.

$$\Delta y_{ir} = \alpha_r + \beta_W S_{ir} + W'_{ir}\gamma + \varepsilon_{ir} \tag{3}$$

Δy_{ir} is the change in municipality i of region r, S_{ir} its own offshore work, α_r a fixed effect for the region and W_{ir} the municipal versions of the controls of Equation (1). The regressions are weighted by municipal population on 1 January 2014. The fixed effect removes everything that a region shares, such as its distance to the oil centres, its coast, its housing market and its trend. What is left is the comparison of a municipality with more offshore workers to its neighbours. Offshore pay is counted where the worker lives, so this comparison shows the loss of pay itself. It cannot show the effect on prices, because neighbouring municipalities share a housing market. As a benchmark I use the direct pay loss, the fall in income if every lost offshore job lost its full pay. In the fourth quarter of 2016 pay in extraction and its services was 1.57 times the average for all industries, so the direct loss is 1.57 times the offshore jobs lost per 100 employed residents.

4.3 Inference

With 86 regions I use heteroskedasticity-robust standard errors and take every p-value and star from the wild bootstrap with the six-point weights of Webb (2023) and 9,999 draws (Cameron et al., 2008; Roodman et al., 2019). The event studies and the within-region regressions cluster by region. Two checks do not rely on large-sample approximations. The first reshuffles offshore work among the regions of the same county 5,000 times and asks how often a reshuffled coefficient lies as far out as the estimated one. Because each county keeps its own mix of exposure, the test also asks whether the effect is only a contrast between the coastal and the inland counties. The second clusters the standard errors of the headline and the county fixed-effects specifications by the 18 counties. The main table also reports Romano–Wolf p-values for the five bust outcomes tested together (Romano and Wolf, 2005, 2016; Clarke et al., 2020).

4.4 Identification

The identifying assumption is that, given the controls, regions with more offshore workers in 2003 would have followed the same path after 2013 as the others had the oil price not fallen. With one national shock, identification comes from the shares (Goldsmith-Pinkham et al., 2020).

Five threats matter. The first is the oil economy next door. Sample regions near Stavanger and Bergen could have been hit through commuting and trade with the oil centres rather than through their offshore workers. The share of a region's workers who commuted to the oil regions in 2003 and

the distance to Stavanger and Bergen measure that contact, and the 58 regions with no onshore oil job at all show the effect without it. The second is the equipment suppliers, whose share is a control. The third is geography. Offshore workers live on the coast, and the weaker krone helped the coastal exporters. The coastal controls, county fixed effects and the permutation within counties address it. The fourth is a trend before the bust. The only years just before it are boom years, so I also use the trend of house prices from 1992 to 2002. The fifth is Oslo, which holds a sixth of the sample's population, had little offshore work and had a price boom of its own. One question the data cannot settle is how far the effect reaches. Offshore work is clustered along the coast, so a region with many offshore workers usually has neighbours with many, and the effect of a region's own offshore workers cannot be told apart from that of its neighbours' (Section 6.4).

5 Results

Table 4 reports the bust across the 86 regions. In the headline specification of column (4), from 2013 to 2017 the regions lost 0.39 offshore jobs per 100 employed residents for each percentage point of offshore work in 2003. Average income from employment fell by 2.2 log points, employment of residents by 1.6 and the average sale price of dwellings by 9.6. The price per square metre of detached houses, which holds the type of dwelling fixed and scales by floor area, fell by 9.7. The dwelling stock and the population did not move. The three effects survive the Romano–Wolf correction for five outcomes, and with standard errors clustered by county the income and price effects keep p-values of 0.002 and below 0.001 and the employment effect 0.053. Across the regions, offshore work ran from 0.08 percent at the 10th percentile to 1.51 percent at the 90th, so a region at the 90th percentile lost about 3 percent more income and about 14 log points more in prices than one at the 10th. Prices fell 4.3 times as much as income.

The controls matter for employment and population and little for income and prices. Without controls, employment fell by 3.4 log points and population by 1.7 per point, because the regions with offshore workers are small and rural. The log population and the 2003 industry shares take employment to 1.6 and population to zero and leave the price at 9.9. The supplier share changes little. The coastal controls of column (5) leave the income effect at 1.9 and raise the price effect to 11.4. Column (6) compares regions within the same county and column (7) keeps the 58 regions with no onshore oil job in 2003. Section 5.1 takes up both.

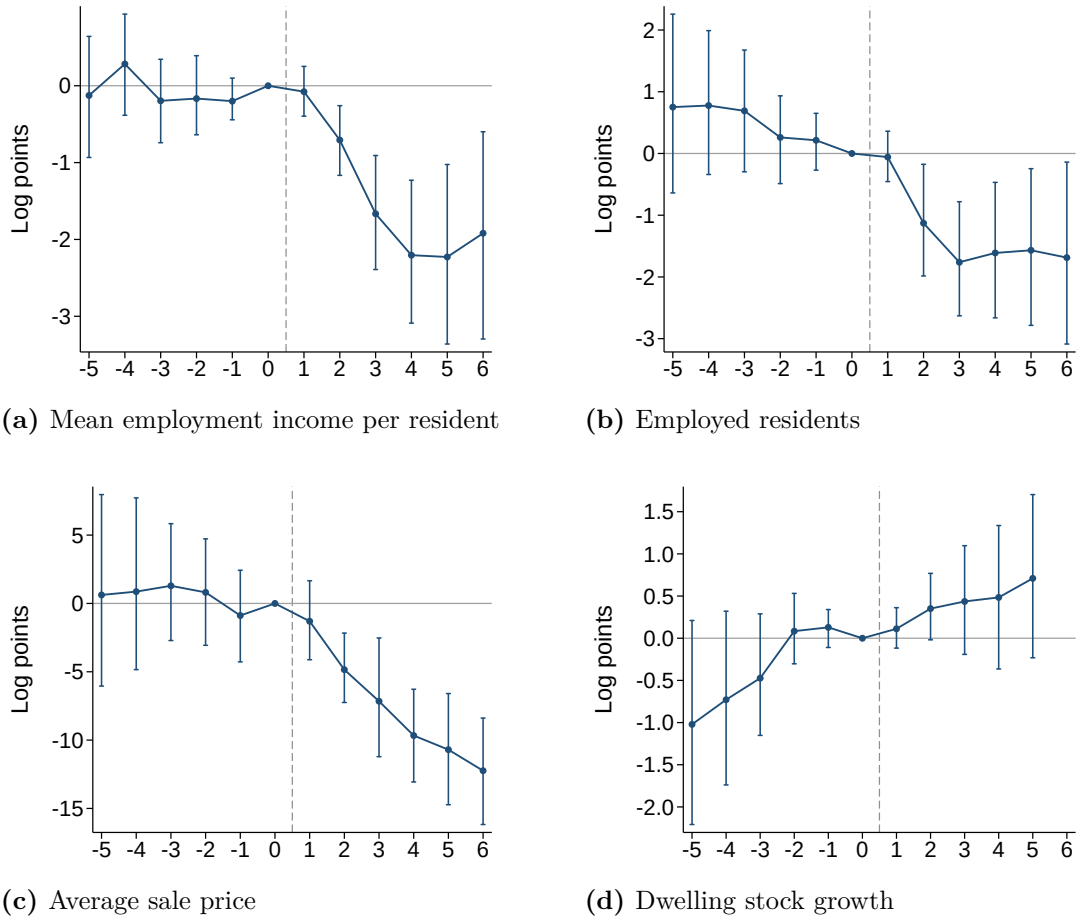
Figure 3 shows the timing. Income and the average sale price were flat relative to 2013 from 2009 to 2012. Prices began to fall in 2014, were 9.7 log points lower per point of offshore work in 2017 (9.6 in Table 4, which weights by the population of 2014) and kept falling to 12.2 in 2019. Income fell from 2015, by 2.2 in 2017, and was still 1.9 lower in 2019. Employment fell in 2015 and 2016 and stayed down. It is also the one series that moved before the bust. Exposed regions lost employment relative to the others by about 0.75 log points per point from 2008 to 2013. No single year can be told from zero, but the drift makes the employment effect the least secure of the three. If the trend of 2009 to 2012 continued, the 2017 effect would keep an interval that just excludes zero

Table 4: The bust across economic regions, 2013–2017

	No controls (1)	Onshore oil (2)	Population, industry (3)	Suppliers (headline) (4)	Coastal (5)	County FE (6)	Clean regions (7)
Panel A. Offshore jobs per 100 employed residents							
Offshore work, 2003	−0.367***	−0.384***	−0.391***	−0.390***	−0.344***	−0.332***	−0.241***
	(0.051)	(0.045)	(0.065)	(0.064)	(0.058)	(0.048)	(0.057)
Adjusted R^2	0.70	0.71	0.78	0.78	0.79	0.84	0.75
Panel B. Mean employment income							
Offshore work, 2003	−2.655***	−2.738***	−2.315***	−2.248***	−1.881***	−1.499	−1.815**
	(0.426)	(0.433)	(0.485)	(0.408)	(0.503)	(0.757)	(0.629)
Adjusted R^2	0.39	0.39	0.51	0.63	0.69	0.72	0.43
Panel C. Employment							
Offshore work, 2003	−3.429***	−4.125***	−1.596**	−1.560***	−2.047***	−1.674**	−1.481
	(0.824)	(0.740)	(0.529)	(0.504)	(0.706)	(0.706)	(0.930)
Adjusted R^2	0.28	0.41	0.67	0.69	0.69	0.74	0.30
Panel D. Average sale price							
Offshore work, 2003	−9.627***	−10.018***	−9.895***	−9.647***	−11.419***	−9.419**	−7.684
	(1.257)	(1.291)	(2.000)	(1.648)	(2.494)	(4.281)	(4.759)
Adjusted R^2	0.27	0.26	0.26	0.35	0.34	0.35	0.12
Panel E. Detached price per square metre							
Offshore work, 2003	−11.339***	−12.951***	−9.921***	−9.658***	−10.139***	−5.122*	−7.866**
	(1.784)	(1.533)	(1.938)	(1.538)	(1.988)	(2.982)	(2.875)
Adjusted R^2	0.44	0.51	0.57	0.65	0.64	0.71	0.36
Panel F. Dwelling stock growth							
Offshore work, 2003	−0.364	−0.180	0.533	0.536	−0.316	−0.165	−1.054
	(0.442)	(0.441)	(0.399)	(0.404)	(0.551)	(0.745)	(0.927)
Adjusted R^2	0.00	0.00	0.19	0.18	0.20	0.37	0.21
Panel G. Population growth							
Offshore work, 2003	−1.680**	−2.143***	−0.247	−0.260	−0.996	−0.776	−0.393
	(0.598)	(0.544)	(0.417)	(0.412)	(0.641)	(0.672)	(0.833)
Adjusted R^2	0.12	0.23	0.58	0.58	0.59	0.69	0.35
Regions	86	86	86	86	86	86	58
Onshore oil controls		✓	✓	✓	✓	✓	✓
Population and industry controls			✓	✓	✓	✓	✓
Supplier share				✓	✓	✓	✓
Coastal controls					✓		
County fixed effects						✓	
Population weights	✓	✓	✓	✓	✓	✓	✓

Note: One observation per economic region. Offshore work is the 2003 share of employed residents whose workplace was the continental shelf, in percentage points, averaged over the region's sample municipalities (population weights of 1 January 2004). Outcomes are population-weighted regional means of the municipal changes from 2013 to 2017 in log points; the dwelling stock and population run from 1 January 2014 to 1 January 2018, and offshore jobs are the change in residents in SN2007 06 and 09.1 per 100 employed residents of 2013. The average sale price uses municipalities with at least ten sales every year from 2003 to 2019; the detached price per square metre (table 06035) covers 82 regions (55 in column 7). Controls are added in turn: (2) the regional onshore oil job share and the regional mean of the residents' onshore oil share; (3) log regional population and the 2003 shares in industry, the primary sector and trade; (4) the 2008 share of residents in fabricated metal products and machinery; (5) the fishing, shipbuilding and water-transport shares, log distance to a heliport and the regional share of petroleum-related jobs; (6) county fixed effects. Column (7) keeps the 58 regions with no onshore oil job in 2003. Weights are regional population. Robust standard errors in parentheses; stars from the wild bootstrap (Webb weights, 9,999 draws), * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. With county clusters, wild cluster p for panels A–G: column (4) <0.001, 0.002, 0.053, <0.001, <0.001, 0.376, 0.677; column (6) 0.002, 0.006, 0.040, 0.046, 0.172, 0.873, 0.246. Column (4): Romano–Wolf p for income, employment, price, stock and population = 0.001, 0.033, 0.001, 0.413, 0.556; permutation within counties (5,000 draws) $p = 0.003$ (income), 0.015 (employment), 0.008 (price), 0.004 (price per square metre).

with the headline controls, from −1.7 to −0.1, and would lose it with the coastal controls (Table 7). The dwelling stock grew a little faster in exposed regions until 2011, and after 2013 no year differs from zero at 5 percent, although if anything the stock kept growing faster. The detached price per square metre is not in the figure, because its years before the bust are not flat. Section 6.2 takes it up.



(a) Mean employment income per resident

(b) Employed residents

(c) Average sale price

(d) Dwelling stock growth

Figure 3: The bust across economic regions, 2008 to 2019

Note: Coefficients on the region's offshore work in 2003 by year, from a regression of the regional change from 2013 on year effects and on offshore work and the controls of column (4) of Table 4, each interacted with year. Each economic region is one observation per year. The regression is weighted by regional population on 1 January 2004. The bars are 95 percent intervals from the wild bootstrap clustered by region (Webb weights, 9,999 draws). The horizontal axis counts years from 2013, so −5 is 2008 and 6 is 2019, and the dashed line falls between 2013 and 2014. Outcomes are 100 times the log difference from 2013, averaged over the region's sample municipalities with population weights. The average sale price uses the municipalities with at least ten sales in every year from 2003 to 2019 (86 regions). Dwelling stock growth is 100 times the log difference of the end-of-year stock from 2013 and ends with the stock of 1 January 2019. The detached price per square metre is shown from 2003 in Figure 7.

5.1 Where the industry is absent

The regions with no onshore oil job in 2003 give the same picture with less precision. In column (7) of Table 4, income fell by 1.8 log points per point of offshore work and the price per square metre

by 7.9. The average sale price fell by 7.7 but cannot be told from zero, because it mixes the types of dwellings sold in thin markets. The slopes in these 58 regions and in the other 28 cannot be told apart.[12] Reshuffling offshore work among the regions of the same county gives p-values of 0.003 for income, 0.015 for employment, 0.008 for the average sale price and 0.004 for the price per square metre in all 86 regions, and of 0.007, 0.012 and 0.055 for income and the two prices in the clean regions. The effect is not a contrast between counties.

Comparing regions within a county is weaker. In column (6), the average sale price falls by 9.4 log points per point, but income by 1.5, which cannot be told from zero with robust standard errors, and the price per square metre by 5.1, which is significant only at 10 percent. Within counties only a sixth of the variation in offshore work is left, and the result rests on one county. Before the supplier share enters the controls, county fixed effects give an income effect of 0.6 and a price effect of 6.0. Leaving out Vest-Agder restores 1.8 and 12.0 (2.5 and 14.5 with the supplier share, Table 6), and leaving out any other county does not. The least exposed region of Vest-Agder is Kristiansand, whose residents held 5.9 supplier jobs per 100 employed in 2013 and lost 2.0 of them by 2017, against 0.9 to 1.4 in the three more exposed regions of the county. There the comparison within the county sets the exposed regions against a region that was hit through its suppliers. I found this after seeing the results, so it is an explanation and not a test. An independent measure of the suppliers, the share of jobs located in fabricated metal products and machinery in 2003, recovers the price effect only in part, 6.9 log points with a p-value below 0.10, and not the income effect (Table 6). Section 6.4 gives a second reason. If the effect reaches the neighbouring regions, a comparison of neighbours understates it.

5.2 Within and between regions

Table 5 sets the two comparisons side by side. Within regions, a municipality with one percentage point more offshore work lost 0.22 more offshore jobs per 100 employed residents than its neighbours. Valued at the pay of the industry, that is a direct loss of 0.34 percent of employment income. Its income fell by 0.68 log points, about twice the direct loss, and by 0.41 in the regions without onshore oil, about the direct loss itself. Neither income coefficient can be told from zero, but both are on the scale of the pay that the offshore jobs carried. Prices did not fall detectably more than in the neighbouring municipalities, with coefficients of 1.7 for the average sale price and 1.1 for the price per square metre. Between regions the direct loss is 0.61 percent and income fell by 2.2 log points, 3.7 times as much. The within-region estimates cannot tell the two ratios apart. A bootstrap over regions puts the difference between the ratio across regions and the ratio of 2.0 within them at 1.7, with a 95 percent interval from -2.5 to 4.9, and at 3.7, from -1.9 to 8.7, in the regions without onshore oil.[13] The income loss beyond the offshore pay therefore rests on the comparison across

[12] With offshore work interacted with an indicator for the 58 regions, the p-values for equal slopes are 0.24 for income, 0.16 for the average sale price and 0.31 for the price per square metre.

[13] Each draw resamples the 86 regions with replacement, each with all of its sample municipalities, and estimates both designs. The 1,999 draws give percentile intervals. The offshore coefficient within regions is negative in every draw.

regions, and the fall in prices is regional by design, since neighbouring municipalities share a housing market. Under the identifying assumption the regional estimates are what the lost jobs and pay did to the regions where they were earned and spent.

Table 5: Within and between regions, bust 2013–2017

	Within regions		Between regions	
	All	Clean	All	Clean
	(1)	(2)	(3)	(4)
Offshore jobs per 100 employed residents	−0.219***	−0.251***	−0.390***	−0.241***
	(0.063)	(0.072)	(0.064)	(0.057)
Direct pay loss (1.57× offshore jobs)	−0.344	−0.395	−0.613	−0.379
	(0.098)	(0.113)	(0.101)	(0.089)
Mean employment income	−0.679	−0.413	−2.248***	−1.815**
	(0.422)	(0.510)	(0.408)	(0.629)
Employment	−0.706	0.269	−1.560***	−1.481
	(0.524)	(0.556)	(0.504)	(0.930)
Average sale price	1.657	0.560	−9.647***	−7.684
	(3.051)	(6.797)	(1.648)	(4.759)
Detached price per square metre	1.119	−2.188	−9.658***	−7.866**
	(1.249)	(2.846)	(1.538)	(2.875)
Observations	375	256	86	58
Observations, average sale price	271	170	86	58
Observations, price per square metre	225	137	82	55
Unit of observation	Municipality	Municipality	Region	Region
Economic-region fixed effects	✓	✓		

Note: Coefficients per percentage point of offshore work in 2003. Columns (1) and (2) compare municipalities within an economic region. They add region fixed effects to the municipal versions of the controls of column (4) of Table 4: the municipal onshore oil measure, log population, the 2003 shares in industry, the primary sector and trade, and the 2008 supplier share. They are weighted by municipal population on 1 January 2014, with standard errors clustered by region and stars from the wild cluster bootstrap. Column (2) keeps the municipalities of the regions with no onshore oil job in 2003. Columns (3) and (4) are columns (4) and (7) of Table 4. The direct pay loss is the percentage of employment earnings that would be lost if every lost offshore job lost its full pay. Pay in extraction and its services was 1.57 times the all-industry average in the fourth quarter of 2016 (Statistics Norway). The row is a benchmark and is not tested. Outcomes, inference and stars otherwise as in Table 4.

5.3 Who lost

The income loss fell on men, who held nine in ten of the offshore jobs. Income per adult man fell by 3.4 log points per point of offshore work and per adult woman by 0.3, which cannot be told from zero. For men the loss splits into fewer men in work, 1.2 log points, and lower pay for those in work, 2.2. The two add up to the loss in income per man.

The offshore jobs lost were 0.39 per 100 employed residents per point, while total employment fell by 1.6. Not all of the difference is a multiplier. The equipment suppliers of the industry are machinery and metal fabrication and the support codes on Statistics Norway's petroleum list. They lost 0.45 jobs per 100, about one for each offshore job.[14] The sample rule removes the oil companies

[14] 1.1 for each offshore job, with a 95 percent interval from 0.5 to 2.3 (Table 13). Tables 11 and 12 report the change

and their service firms but not their suppliers. Counted by place of work, the supplier jobs located in the sample regions fell by 0.23 per 100 employed residents per point, about half of the residents' loss, so part of the hit was on local firms and part on supplier firms that residents commuted to. The split is imprecise.[15] The price effect does not depend on the supplier share. With offshore work interacted with it, the average sale price falls by 9.3 log points per point at the mean supplier share and by 8.7 at the 10th percentile, and the interaction cannot be told from zero ($p = 0.72$). The rest of the local economy lost 0.73 jobs per 100. That is 1.9 for each offshore job, but the interval runs from -1.2 to 4.9, and without the rest of manufacturing the ratio is 0.9. Outside oil and the suppliers the losses were in hotels and restaurants, 0.24 jobs per 100, and in finance. No industry gained except agriculture, forestry and fishing. Valued at the pay of the oil industry, the offshore jobs lost explain 27 percent of the fall in employment income.[16]

Unemployment insurance cushioned part of the loss, but other income did not make up for it on net. Per point of offshore work, employment income per adult fell by 2.2 log points and gross income, which adds benefits, pensions and capital income for the same people, by 1.7. Employment income was about 70 percent of gross income in 2013, so the lost pay alone would have lowered gross income by about 1.6. The median household income after tax fell by 1.5. An offshore worker who lost the job received 62.4 percent of previous earnings for up to two years (Section 2.3). Per point of offshore work, the number of residents who received unemployment benefits during 2016 was 1.3 per 100 employed residents higher than in other regions. That is about four recipients for every five jobs lost and more than three for every offshore job lost, so the benefits reached workers outside oil too. The benefits paid were about 53 dollars per resident higher in 2016.[17] A third of the lost offshore jobs had come back by 2019. Prices fell all the same, and they fell on an unchanged volume of sales (Section 6.5).

5.4 How far should prices fall?

With the stock of dwellings and the population fixed, a fall in demand cannot be met by building less or by fewer residents, so it shows up in prices (Glaeser and Gyourko, 2005). If housing demand depends on permanent income with elasticity η_y and on the price with elasticity η_p, a fixed stock clears when the price moves by η_y/η_p times the change in permanent income. Norwegian house-price equations estimate that ratio directly, as the long-run elasticity of house prices with respect to income with the housing stock held fixed. It is 1.66 in Jacobsen and Naug (2005), 1.62 in Statistics Norway's macroeconomic model (Boug and Dyvi, 2008) and 1.69 in Anundsen and Jansen (2013)

in employed residents by industry of workplace, for the industry groups and for the petroleum-related industries and the manufacturing divisions, and Table 14 repeats the decomposition on the supply-chain samples.

[15] Statistics Norway table 13470, employed persons by place of work, in the same industries, per 100 employed residents of 2013. The standard error is 0.22, and the ratio of the two coefficients is 0.52 with a standard error of 0.38.

[16] The direct loss is $1.57 \times 0.39 = 0.61$ percent of employment earnings, against a fall of 2.25 log points in employment income per adult. It counts every lost job as a full year of pay lost and leaves out pay cuts for those who kept their jobs.

[17] 444 kroner, converted at Norges Bank's average rate for 2016 of 8.40 kroner per dollar. By 2017 the difference was 91 kroner and could not be told from zero.

with household debt held fixed.

Per point of offshore work, prices fell by 9.6 log points and income by 2.2, 4.3 times as much. With a ratio of 1.66, a lasting loss of income of the 2017 size lowers prices by 3.7 log points, about two-fifths of the fall. The ratio alone cannot test the benchmark, because a large enough expected loss would match any ratio above 1.66. What the forecasts of the time add is that the loss was expected to last. In every report from December 2015 to December 2016, Norges Bank put petroleum investment in 2018 and 2019 at 63 to 69 percent of its 2013 level, with no return to it (Norges Bank, 2015, 2016a,b,c). Statistics Norway expected employment linked to petroleum to fall from 232,100 in 2013 to about 182,500 in 2016 (Statistics Norway, 2016), and Blomgren et al. (2016) expected 50,000 direct petroleum jobs to be lost by 2018 and 13,000 to return by 2020. The government's long-term perspective of 2017 expected petroleum-related employment to be nearly halved by 2040 (Finansdepartementet, 2017). Only the industry's own outlook of October 2015, at an oil price of 70 dollars, expected investment to recover by 2019 (Norsk olje og gass, 2015). These forecasts expected the loss to last at about its size of 2017, not to grow to 2.6 times that size, which is what a market that priced the news at once would need to explain the whole fall. Two things account for part of the gap. The dwelling stock grew by 0.5 log points more per point of offshore work, which in the same equations lowers prices by about 0.9. And prices kept falling to 2019 while income recovered part of the way and a third of the lost offshore jobs came back (Figure 3). A market that priced the news at once would not keep falling, so part of the fall is slow adjustment rather than repricing. Measured by the median household income after tax, which fell by 1.5 log points, the ratio is 6.3, so the gap is larger, not smaller.

The ratio does not depend on how many offshore workers the register misses. Counting the hours worked offshore as full-time jobs, there were 1.2 to 1.4 offshore workers for each resident registered on the shelf between 2003 and 2014 (Petroleumstilsynet, 2016; Fjelldal and Blomgren, 2023). The undercount makes the effect of one true percentage point of offshore work smaller than the coefficients, but it scales income and prices alike. Glaeser and Gyourko (2005) also predict that prices respond more to falling than to rising demand, and Notowidigdo (2020) finds no such asymmetry across US cities. The benchmark does not rest on the asymmetry, only on the fixed stock.

6 Robustness

Table 6 changes one element of the headline specification at a time, with three measures of the price in every row. The firm-side coastal controls, which leave out water transport and the distance to a heliport, give -1.9 for income and -10.7 for the price. Without Oslo, which holds a sixth of the sample's population, the effects are -2.1 and -9.7. Without population weights they are -1.4 and -8.4, and a median regression gives -1.7 and -8.1. Measuring the suppliers by the jobs located in the region in 2003 instead of the residents in 2008 gives -2.4 and -10.2. The detached price per square metre and the average price smoothed over 2011–13 and 2015–19 keep the sign and the

significance of the average sale price in every row except those with county fixed effects, where the price per square metre is weaker (Section 5.1).

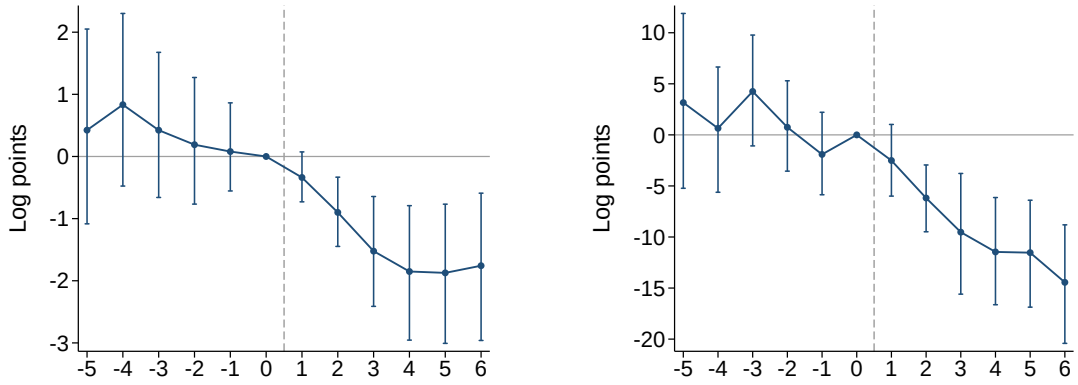
Table 6: The regional effect under alternative specifications, bust 2013–2017

	Income	Employment	Average sale price	Price per square metre	Smoothed sale price
	(1)	(2)	(3)	(4)	(5)
Headline	−2.248***	−1.560***	−9.647***	−9.658***	−9.148***
	(0.408)	(0.504)	(1.648)	(1.538)	(1.319)
Firm-side coastal controls	−1.949***	−1.648**	−10.705***	−10.555***	−10.264***
	(0.420)	(0.600)	(2.050)	(1.640)	(1.550)
Without Oslo	−2.102***	−1.409**	−9.736***	−8.686***	−9.370***
	(0.407)	(0.489)	(1.783)	(1.407)	(1.444)
Unweighted	−1.353**	−0.897**	−8.441***	−6.249***	−7.040***
	(0.477)	(0.405)	(2.862)	(1.636)	(1.646)
Median regression	−1.714**	−0.937	−8.145**	−5.381**	−7.312***
	(0.756)	(0.640)	(3.868)	(2.677)	(2.010)
Supplier jobs located in 2003	−2.409***	−1.644***	−10.206***	−10.092***	−9.633***
	(0.454)	(0.523)	(1.860)	(1.840)	(1.658)
County FE, supplier jobs 2003	−0.817	−1.163*	−6.895*	−3.689	−3.169
	(0.719)	(0.659)	(4.033)	(2.767)	(2.745)
County FE, without Vest-Agder	−2.465**	−2.144**	−14.516**	−8.201*	−9.246**
	(0.953)	(1.041)	(6.425)	(4.679)	(4.236)
Area exposure within 100 km	−2.705***	−0.853	−10.078***	−9.582***	−9.955***
	(0.510)	(0.677)	(2.268)	(2.273)	(1.758)
Area exposure within 150 km	−3.031***	−0.199	−8.619**	−9.694**	−10.073***
	(0.488)	(0.771)	(2.827)	(2.710)	(2.256)
Counties (18)	−2.367*	0.066	−7.524**	−7.644**	−7.613**
	(0.924)	(1.011)	(2.252)	(2.771)	(2.827)

Note: Each row changes one element of column (4) of Table 4, which is repeated in the first row. Column (4) uses the 82 regions with a detached price per square metre in 2013 and 2017. Column (5) is the log change in the mean of the average sale price from its 2011–13 mean to its 2015–19 mean. The firm-side controls are the coastal controls without water transport and heliport distance. The median regression reports robust standard errors and a t-test, with no bootstrap. The supplier jobs located in 2003 are the share of jobs located in the region in fabricated metal products and machinery (SN2002 28 and 29; Statistics Norway table 08545). The Vest-Agder row drops the county's four regions. The exposure of the area within k km is the population-weighted mean of offshore work over the region and every sample region whose centre lies within k km. The county row has one observation per county, with offshore work, the supplier share and the onshore oil share as county means and the log of county population as controls. Otherwise outcomes, weights, inference and stars are as in Table 4. On the 211 municipalities (81 regions) with an average sale price in every year from 1992, the specification of column (4) of Table 4 gives a 2013–2017 price effect of −10.1; with the 1992–2002 trend extrapolated its 95 percent interval is $[-13.1, -4.8]$ (Rambachan and Roth 2023, $M = 0$).

6.1 The coast

In the earlier municipal version of this paper the coastal controls took two-fifths from the income effect and a quarter from the price effect (Table 10). Across regions they take a sixth from income and add to the price (column (5) of Table 4), and the event studies keep their shape with them (Figure 4). Two of the coastal controls also carry part of the oil economy. The share of residents in water transport predicts the loss of transport jobs in the bust, 0.16 jobs per 100 employed residents per point of the share, and faster growth before it, which fits the crews of the supply ships. The distance to a heliport does not pick up oil workers that the shelf count misses, but regions far from



(a) Income, coastal controls

(b) Average sale price, coastal controls

Figure 4: Income and price with the coastal controls, 2008 to 2019

Note: The event studies of Figure 3 for the mean employment income per resident and the average sale price, with the coastal controls of column (5) of Table 4 interacted with year. The panel starts in 2003 (income 2005). Intervals, weights and axes as in Figure 3.

a heliport had slower income growth from 2009 to 2013, so it belongs among the controls. Holding it fixed alone leaves the effects at −2.3 for income and −9.9 for the price.

6.2 Trends before the bust

Table 7 asks how far a trend before 2013 could continue after it before the 2017 effect loses significance (Rambachan and Roth, 2023). With the trend of 2009 to 2012 as the reference, the price effect survives a straight continuation of the trend with the headline and the firm-side controls, and the income and employment effects with the headline controls, employment only just. With the full coastal controls none does. Those four years are boom years, though, and a slope fitted to four points is extrapolated over four to six. The price series reaches back to 1992 in 211 sample municipalities in 81 regions, which gives eleven years before the boom. With the trend of 1992 to 2002 as the reference, the bust from 2013 to 2017 keeps an interval that excludes zero in all three specifications, from $[-13.1, -4.8]$ with the headline controls to $[-13.7, -3.1]$ with the coastal controls, and the trend may change its slope by less than 0.1 log points a year (0.2 with the firm-side controls) before the interval reaches zero. The coefficients of 1992 to 2002 are jointly zero, with p-values of 0.54 and 0.23 with the headline and the coastal controls (Figure 5). Across municipalities, the earlier version of this paper reported no trend in the average sale price from 1992 to 2002. A joint test of zero coefficients has little power against a steady drift, and the municipal coefficients do drift down by 0.45 log points a year (standard error 0.20). Across regions the drift is 0.32 (0.24). The pre-boom bounds are for the average sale price.

The detached price per square metre does not pass them. On the 75 regions with a price in every year from 2008, it was 5.2 log points per point of offshore work above its 2013 level in 2009 and drifted down to it by 2012. With that trend continuing, the interval for 2017 runs from −9.2 to 0.0. In 56 regions the series reaches back to 2003 (Figure 7). There it was flat from 2003 to 2006,

Table 7: Sensitivity of the 2017 effect to the trend before the bust

	Average sale price			Employment income		Employment	
	Headline (1)	Firm-side (2)	Coastal (3)	Headline (4)	Coastal (5)	Headline (6)	Coastal (7)
Panel A. Reference: the trend of 2009–2012							
Coefficient, 2017	−9.67	−10.73	−11.46	−2.20	−1.85	−1.61	−2.14
95% interval	[−13.0, −6.4]	[−14.8, −6.6]	[−16.4, −6.5]	[−3.0, −1.4]	[−2.8, −0.9]	[−2.6, −0.6]	[−3.5, −0.8]
Trend continues, $M = 0$	[−12.9, −0.9]	[−15.9, −0.7]	[−15.2, 1.6]	[−2.6, −0.6]	[−2.1, 0.7]	[−1.7, −0.1]	[−2.0, 0.2]
Breakdown M	0.12	0.12	0.00	0.20	0.00	0.05	0.00
Breakdown $\bar{M}$	0.75	0.75	0.50	0.75	0.50	0.50	0.50
Panel B. Reference: the trend of 1992–2002							
2013–17, base 2003	−10.13	−10.95	−11.10				
Trend continues, $M = 0$	[−13.1, −4.8]	[−15.6, −6.1]	[−13.7, −3.1]				
Breakdown M	0.100	0.200	0.100				

Note: Bounds of Rambachan and Roth (2023) for the region event study of Equation (2). Panel A uses the 2009–2012 coefficients as the pre-period and the 86 regions. Rows labelled M use the smoothness restriction, which lets the trend continue after 2013 with its slope changing by at most M a year; $M = 0$ is a straight continuation, and its row gives the 95 percent interval. The row labelled $\bar{M}$ uses the relative-magnitudes restriction, which bounds each yearly violation after 2013 by $\bar{M}$ times the largest yearly movement before it. Panel B uses base year 2003, the 1992–2002 coefficients as the pre-period and the 211 municipalities in 81 regions with an average sale price in every year from 1992 to 2019, and the target is the change of the coefficient from 2013 to 2017. Income and employment start in 2005 and 2008, so they have no pre-boom period. The breakdown value is the smallest M or $\bar{M}$ on the grid at which the 95 percent interval contains zero; the grid of Panel B is 0, 0.025, 0.05, 0.1 and 0.2. Covariances are clustered by region. Headline, firm-side and coastal are the controls of column (4) of Table 4, with the firm-side coastal controls of Table 6 and the coastal controls of column (5).

rose by 4.6 log points over 2007 to 2010 relative to that level, and was back at it in 2012, with a difference of −0.3 (standard error 2.4). Measured against the level of 2003 to 2006, the bust to 2017 is −10.8 (2.7). With the coastal controls the rise is smaller and cannot be told from zero, but by 2012 the price was already 5.4 log points below its level of 2003 to 2006 ($p = 0.06$), so in that version the price drifted down before the bust. I tried the controls that could take the rise out. The distance to a large city, which flattened the years before the bust for the average sale price in the municipal version of this paper, leaves it unchanged. Statistics Norway's centrality index halves it. The coefficients of 2009 to 2012 are then jointly zero ($p = 0.27$), 2009 is still significant at 10 percent, and the bust in 2017 is −7.8. I chose that control to clear the pre-trend, so it is a check and not a specification. On timing the price per square metre is therefore the weaker of the two measures. The paper uses it for the long differences, where it checks that the fall in the average sale price is not a change in the dwellings sold.

6.3 The boom and the national cycle

Two threats concern the years before the bust, and Table 8 answers both. If the bust were the mirror image of the boom, one coefficient on national offshore employment would describe each outcome over the whole cycle. For income it does not. Income rose with offshore employment in the early boom and fell with it in the bust, but it did not move from 2009 to 2012 while offshore employment kept growing, and the proportional path is rejected with a p-value of 0.005. For prices the coefficients of the boom years are small relative to their standard errors, −2.7 (2.7) for 2003

Table 8: One coefficient on national offshore employment, 2003–2017

	Mean employment income		Average sale price	
	(1)	(2)	(3)	(4)
Whole cycle	−1.38***	−1.51***	−3.88	−5.87**
	(0.40)	(0.33)	(2.20)	(1.82)
2003–2008	−1.54**	−0.27	−2.66	3.50
	(0.63)	(0.64)	(2.75)	(4.15)
2009–2012	−0.08	0.63	1.24	4.25
	(0.42)	(0.51)	(3.24)	(3.39)
2014 onward	−1.96***	−2.41***	−8.93***	−10.83***
	(0.37)	(0.46)	(1.65)	(1.95)
Offshore work × national house prices		−0.007		−0.063**
		(0.008)		(0.026)
Proportional path, p-value	0.005	0.003	0.013	0.016
National house-price term		✓		✓

Note: Coefficients β of Equation (4), per percentage point of offshore work in 2003 and per unit of the 2013–2017 fall in national offshore employment, on the region panel of Figure 3 from 2003 (income 2005) to 2017. The rows for periods estimate β separately for 2003–2008, 2009–2012 and 2014–2017 in one regression. Columns (2) and (4) add offshore work times the national house-price index (κ). The last row is the p-value of the joint test that the event-study coefficients equal βZ_t. Standard errors clustered by region in parentheses; stars from the wild cluster bootstrap. Controls, weights and inference otherwise as in Figure 3.

to 2008 and 1.2 (3.2) for 2009 to 2012, so the fall after 2014 is not the late boom unwinding. The second threat is that exposed regions follow the national house-price cycle less closely. They do, with a coefficient of −0.06 on the national index, and holding the national cycle fixed makes the bust price effect larger, −10.8 instead of −8.9.

$$\bar{y}_{rt} = \lambda_t + \beta\, \bar{S}_r Z_t + \kappa\, \bar{S}_r h_t + \sum_{k \neq 2013} \mathbf{1}[t = k]\, X_r' \gamma_k + u_{rt},$$
$$Z_t = \frac{N_{2013} - N_t}{N_{2013} - N_{2017}} \tag{4}$$

N_t is the number of residents working offshore in year t, so Z_t is the national shortfall of offshore work below its 2013 level in units of its fall from 2013 to 2017. It is zero in 2013, one in 2017 and positive in the boom years, when fewer residents worked offshore than in 2013. h_t is 100 times the log of the national price index for existing dwellings relative to 2013, and κ measures how much less closely exposed regions follow it. I also estimate β separately for 2003 to 2008, 2009 to 2012 and the bust, and test whether the year-by-year coefficients of Equation (2) equal βZ_t. The controls, weights and clustering are those of Equation (2), and the panel runs to 2017.

6.4 The reach of the effect

The oil economy next door does not explain the effect. The share of a region's workers who commuted to the oil regions in 2003 ran from 0.8 to 21 percent, and it is unrelated to offshore work. Interacted with offshore work it has a coefficient of 0.33 (standard error 0.25) for the price and −0.04 (0.07)

for income, and at the least connected region the effects are −10.4 and −2.3.[18] How far the effect reaches is a separate question. Offshore work is clustered along the coast. The offshore work of the sample regions within 100 km of a region correlates 0.75 with its own, so a region's own offshore workers and those of its neighbours cannot be told apart. Measuring offshore work over a wider area does not change the price effect. Per point of offshore work in the region and the sample regions within 50, 100 and 150 km, the price falls by 9.4, 10.1 and 8.6 log points, and across the 18 counties by 7.5. The income effect grows with the area, from 2.4 to 3.0, and the employment effect fades. The effect is therefore at least regional and may extend to the surrounding area. If it does, a comparison of neighbouring regions understates it.[19]

6.5 Further checks

The fall in prices did not come with more sales. Per point of offshore work, the number of free-market sales rose by 3 log points from 2013 to 2014 and by 5 from 2013 to 2017, and sales per 1,000 dwellings by at most 2.1 in any year. None of these changes can be told from zero, so prices fell on an unchanged volume (Figure 8). The average sale price is not adjusted for the dwellings sold, and in thin markets one year's sales can move it a great deal.[20] This is why every table also reports the detached price per square metre. In 2008 and 2009 the earnings of men in exposed regions held up relative to the years around them, and the detached price per square metre rose with them. With the coastal controls the price rise disappears (Table 19). Registered unemployment rose by 0.5 percentage points per point of offshore work by 2016 and was back at its 2013 level by 2018, but with the coastal controls the rise is a quarter as large and cannot be told from zero (Figure 6).

The 2015 change of source in the employment register is a level shift of about 3.5 percent that does not follow offshore work. Statistics Norway published the 2015 count of employees by workplace municipality both on the new source and adjusted to the old one. The gap between the two is unrelated to exposure, in total, outside mining and manufacturing, and in public administration, education and health. On the adjusted series, jobs located in exposed municipalities fell by 0.8 per point of exposure from 2014 to 2015 and by 1.0 on the unadjusted series. The fall in the year of the break is therefore the shock and not the break.[21]

Offshore work in 2013 was 1.22 times the 2003 level across regions and correlates 0.95 with it. On the 2013 share the effects are −1.7 for income, −0.9 for employment and −6.8 for the price, a quarter to two-fifths smaller (Table 17). The distance to the nearest large city and Statistics

[18] The commuting share is the share of a region's employed residents whose workplace in 2003 was in one of the regions left out by the sample rule. It correlates −0.06 with offshore work across regions. With the commuting share and the log distance to Stavanger or Bergen as controls the effects are −2.4 for income and −9.6 for the price.

[19] The detached price per square metre falls by 9.4, 9.6 and 9.7 log points per point of offshore work within 50, 100 and 150 km and by 7.6 across counties (Table 6 reports 100 and 150 km and the counties).

[20] In Kvinesdal, one of the most exposed municipalities, the average sale price rose by 24 log points from 2013 to 2014 on an unchanged number of sales, after apartments came on the market, while the detached price per square metre fell by 2 log points from 2013 to 2017.

[21] Table 15. The test is on municipalities, the unit of the break. The adjusted count carries 2014 forward with the change from the first quarter of 2015 to the first quarter of 2016 by industry group and workplace. It is therefore a bridge and not an observation. It counts employees only.

Norway's centrality index change little (Table 18). Before the boom, offshore work in 2003 does not predict population growth in the 1980s and 1990s or the change in employment and registered unemployment from 2000 to 2003 in the 86 regions. One of the 24 coefficients across the samples has a p-value below 0.05, the change in registered unemployment in the 58 regions without onshore oil (Table 16).

7 Conclusion

The 2014 oil price bust is usually read off the oil regions, where the firms, the workers and the landowners were hit together. Norway's offshore rotation separates them. Offshore workers lived all over the country, and the regions with almost no oil firms received the bust mostly as lost pay. Across 86 such regions, each percentage point of employed residents working offshore in 2003 cost about 2 percent of employment income and 9 percent of house prices between 2013 and 2017, while the dwelling stock and the population did not move. Within regions, the income estimates are imprecise but close to the offshore pay that was lost. The loss fell on men. Residents lost about one job at the oil suppliers for each offshore job, partly in the regions themselves, so the shock is to residents' jobs in the oil value chain. The price effect is about as large where few residents worked for the suppliers. The rest of the local economy lost about two jobs for each offshore job, an estimate that cannot be told from zero.

Two things follow. A shock that arrives through residents' jobs and pay is enough to move house prices, and it moves them far more than it moves income. Prices fell about four times as much as income per point of exposure. Norwegian estimates of housing demand put the long-run response of prices to a lasting loss of income at about 1.7 times the loss when the stock of dwellings is fixed, so a lasting loss of the 2017 size, which forecasters expected, explains about two-fifths of the fall. Prices kept falling after income began to recover, so much of the rest looks like slow adjustment. The point estimate of the local multiplier, once the suppliers are set aside, is about two, close to estimates for tradable jobs elsewhere, but its interval includes zero.

The estimates have limits that the paper states. The comparison is between regions, and offshore workers cluster along the coast, so the effect of a region's own offshore workers cannot be told from that of its neighbours', and comparisons of regions within a county are weak. Employment and the price per square metre of detached houses were not flat in exposed regions before the bust, and the employment effect survives a continuation of its earlier trend only with the headline controls. With the coastal controls, neither the income nor the price effect survives a continuation of the trend of 2009 to 2012, although the price effect survives the trend of 1992 to 2002. Within regions the income estimates cannot tell the lost pay from its regional multiple. Public municipal data cannot follow a laid-off worker to a new job or to the sale of a house. Register data on individuals would show whether the workers who stayed took lower-paid jobs or waited on benefits, and whether the dwellings that were sold were theirs. Until then the regional aggregates give the size of the response. Where the industry is absent, its wages still set the price of houses.

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A Offshore work by municipality

Table 9: The municipalities with the most and the least offshore work, 2003

Municipality	Economic region	Employed residents	Working offshore	Share (%)
Panel A. The ten with the most				
Tysnes	Sunnhordland	1,271	54	4.25
Fedje	Bergen	282	11	3.90
Kvinesdal	Flekkefjord	2,617	86	3.29
Bamble	Skien/Porsgrunn	6,638	201	3.03
Farsund	Lyngdal/Farsund	4,391	123	2.80
Bømlo	Sunnhordland	5,051	137	2.71
Lyngdal	Lyngdal/Farsund	3,372	83	2.46
Vega	Brønnøysund	629	15	2.38
Tvedestrand	Arendal	2,614	62	2.37
Sauda	Haugesund	2,290	54	2.36
All 375: p10 0.00, p25 0.10, median 0.28, p75 0.75, p90 1.33 40 at zero; population-weighted mean 0.41				
Panel B. The ten with the least, among municipalities with any				
Gran	Hadeland	6,539	1	0.02
Rakkestad	Fredrikstad/Sarpsborg	3,626	1	0.03
Sørum	Lillestrøm	6,659	2	0.03
Nordre Land	Gjøvik	3,251	1	0.03
Vadsø	Vadsø	3,083	1	0.03
Sel	Nord-Gudbrandsdalen	2,965	1	0.03
Lier	Drammen	11,156	4	0.04
Gjerdrum	Lillestrøm	2,629	1	0.04
Fet	Lillestrøm	4,998	2	0.04
Øyer	Lillehammer	2,459	1	0.04

Note: Residents whose workplace was the continental shelf in the fourth quarter of 2003 as a percentage of employed residents. The sample is the 375 municipalities where at most 0.5 percent of employed residents held onshore oil jobs in 2003, in the municipality or by commuting (Section 3.2). Panel A ends with the distribution of the share across the 375. Forty municipalities had no offshore workers. Nine of them are in Finnmark and six each in Hedmark, Nordland and Trøndelag. The largest, Søndre Land, had 2,692 employed residents. Panel B lists the ten lowest shares among the rest.

B The municipal design

Table 10: The bust across municipalities, 2013–2017

	No controls (1)	Onshore oil (2)	Baseline (3)	Coastal (4)	Distance (5)	Region FE (6)	All 421 (7)
Panel A. Mean employment income							
Offshore work, 2003	-2.293^{***}	-2.340^{***}	-1.932^{***}	-1.200^{***}	-1.139^{***}	-0.717^{*}	-1.708^{***}
	(0.370)	(0.385)	(0.385)	(0.372)	(0.346)	(0.410)	(0.259)
Adjusted R^2	0.25	0.25	0.35	0.45	0.45	0.63	0.67
Observations	375	375	375	375	375	375	421
Panel B. Employment							
Offshore work, 2003	-2.998^{***}	-3.538^{***}	-1.378^{***}	-1.355^{***}	-1.056^{***}	-0.705	-0.823^{***}
	(0.705)	(0.653)	(0.386)	(0.399)	(0.359)	(0.523)	(0.297)
Adjusted R^2	0.19	0.27	0.55	0.56	0.57	0.68	0.54
Observations	375	375	375	375	375	375	421
Panel C. Average sale price							
Offshore work, 2003	-7.718^{***}	-7.725^{***}	-7.142^{***}	-5.158^{**}	-4.127^{**}	1.643	-5.040^{***}
	(1.117)	(1.295)	(1.444)	(1.726)	(1.699)	(2.999)	(1.116)
Adjusted R^2	0.12	0.12	0.13	0.16	0.18	0.31	0.34
Observations	271	271	271	271	271	271	312
Panel D. Dwelling stock growth							
Offshore work, 2003	-0.341	-0.245	0.226	-0.287	-0.310	0.035	0.453
	(0.375)	(0.399)	(0.302)	(0.297)	(0.317)	(0.501)	(0.315)
Adjusted R^2	0.00	0.01	0.12	0.15	0.15	0.37	0.12
Observations	375	375	375	375	375	375	421
Panel E. Population growth							
Offshore work, 2003	-1.505^{**}	-1.893^{***}	-0.284	-0.734^{**}	-0.515^{*}	-0.389	0.016
	(0.491)	(0.438)	(0.272)	(0.296)	(0.279)	(0.327)	(0.264)
Adjusted R^2	0.08	0.16	0.47	0.49	0.50	0.65	0.40
Observations	375	375	375	375	375	375	421
Onshore oil controls		✓	✓	✓	✓	✓	✓
Population and industry controls			✓	✓	✓	✓	✓
Coastal controls				✓	✓		
Distance to a large city					✓		
Economic-region fixed effects						✓	
Onshore oil municipalities kept							✓
Population weights	✓	✓	✓	✓	✓	✓	✓

Note: The design of the earlier version of this paper: each municipality is one observation. Coefficients per percentage point of offshore work in 2003 in the 375 municipalities of the headline sample (Section 3.2); the price is on the 271 municipalities of the balanced panel. Column (2) adds the two onshore oil measures of Section 3.4 and column (3) the log population and the 2003 industry shares, without the supplier share. Column (4) adds the coastal controls and column (5) the distance to the nearest large city. Column (6) keeps the controls of column (3) but replaces the regional onshore measure by economic-region fixed effects; it is the within-region comparison of Table 5 without the supplier share. Column (7) keeps the 46 municipalities that the sample rule leaves out, with the controls of column (3); its price is on the 312 municipalities of the balanced panel. Standard errors clustered by economic region in parentheses, stars from the region wild bootstrap with * $p < 0.10$, ** $p < 0.05$ and *** $p < 0.01$. In column (3) the Romano–Wolf p-values over six outcomes, these five and the change in registered unemployment, are 0.002, 0.017, 0.002, 0.553 and 0.553.

C Employment by industry

Table 11: Employed residents by industry of workplace, change per 100 employed residents

	2013–17 (1)	2013–19 (2)	2015–19 (3)
All industries	−1.57***	−1.61*	−0.54
	(0.49)	(0.71)	(0.51)
Oil and gas, mining and quarrying (05–09)	−0.35***	−0.24**	−0.16*
	(0.07)	(0.09)	(0.08)
All other industries	−1.22**	−1.38*	−0.39
	(0.46)	(0.70)	(0.54)
Agriculture, forestry and fishing (01–03)	0.23**	0.30**	0.16*
	(0.09)	(0.11)	(0.09)
Manufacturing (10–33)	−0.79**	−0.86***	−0.54***
	(0.27)	(0.26)	(0.17)
Electricity, water and waste (35–39)	0.00	−0.02	−0.03
	(0.03)	(0.04)	(0.03)
Construction (41–43)	−0.03	−0.07	−0.02
	(0.13)	(0.18)	(0.16)
Trade (45–47)	−0.14	−0.19	−0.01
	(0.14)	(0.16)	(0.13)
Transport and storage (49–53)	−0.08	−0.08	0.08
	(0.07)	(0.11)	(0.08)
Hotels and restaurants (55–56)	−0.24***	−0.23**	−0.07
	(0.07)	(0.09)	(0.07)
Information and communication (58–63)	−0.01	−0.09	−0.07
	(0.07)	(0.11)	(0.09)
Finance and insurance (64–66)	−0.07*	−0.10*	−0.04
	(0.03)	(0.05)	(0.04)
Real estate, professional and technical services (68–75)	−0.14	−0.21	−0.16
	(0.08)	(0.13)	(0.11)
Business services (77–82)	0.03	0.05	0.12
	(0.07)	(0.11)	(0.09)
Public administration (84)	0.10	0.03	−0.08
	(0.07)	(0.08)	(0.07)
Education (85)	0.02	0.12	0.11
	(0.11)	(0.17)	(0.13)
Health and social work (86–88)	−0.08	0.09	0.19
	(0.15)	(0.20)	(0.16)
Other services (90–99)	0.00	−0.08	−0.02
	(0.05)	(0.06)	(0.06)
Unspecified	−0.02	−0.04*	0.01
	(0.04)	(0.02)	(0.04)

Note: Coefficients per percentage point of offshore work in 2003 across the 86 regions of Table 4. The outcome is the change in the number of employed residents whose workplace is in the industry, from the fourth quarter of the first year to the fourth quarter of the last, summed over the region's sample municipalities and divided by its employed residents in 2013, times 100. The data are Statistics Norway's table 13470 by place of residence, with the SN2007 groups in parentheses. The register changed source in 2015, so columns (1) and (2) contain the break and column (3) lies inside the new source. Controls, weights, inference and stars as in column (4) of Table 4.

Table 12: Employed residents in the petroleum-related industries and in manufacturing, change per 100 employed residents

	2013–17 (1)	2013–19 (2)	2015–19 (3)
Petroleum-related industries, Statistics Norway's list	-0.37^{**}	-0.26	-0.14
	(0.14)	(0.15)	(0.09)
Extraction and support services (06, 09.1)	-0.39^{***}	-0.27^{***}	-0.16^{*}
	(0.06)	(0.09)	(0.08)
Platforms, ships and ship repair (30.11, 33.15)	0.06	0.04	0.01
	(0.13)	(0.13)	(0.10)
Pipelines, supply vessels and terminals	-0.03	-0.03	0.02
	(0.05)	(0.07)	(0.06)
Manufacturing outside the list	-0.85^{***}	-0.90^{***}	-0.55^{***}
	(0.23)	(0.24)	(0.16)
Machinery and equipment (28)	-0.38^{***}	-0.41^{***}	-0.37^{***}
	(0.09)	(0.10)	(0.11)
Fabricated metal products (25)	-0.09	-0.14	-0.08
	(0.07)	(0.08)	(0.07)
Computer, electronic and optical products (26)	-0.14	-0.14	-0.08
	(0.11)	(0.09)	(0.05)
Chemicals (20)	-0.02	-0.05	-0.02
	(0.04)	(0.04)	(0.02)
Ships, boats and other transport equipment (30)	-0.03	-0.05	-0.05
	(0.14)	(0.14)	(0.09)

Note: As Table 11. The petroleum-related industries are Statistics Norway's list (five-digit codes 06, 09.1, 30.111, 30.113, 30.115, 30.116, 33.15, 43.13, 49.5, 50.204, 52.215 and 52.223); manufacturing outside the list is divisions 10–33 less the list's manufacturing codes.

Table 13: Jobs lost for each offshore job lost, with the suppliers purged

	2013–17 (1)	2013–19 (2)	2015–19 (3)
Panel A. Change in employed residents, per 100 employed residents in 2013			
Offshore jobs: extraction and support services (06, 09.1)	-0.39^{***}	-0.27^{***}	-0.16^{*}
	(0.06)	(0.09)	(0.08)
Suppliers: the list's other codes, divisions 25 and 28	-0.45^{***}	-0.53^{***}	-0.42^{***}
	(0.13)	(0.13)	(0.15)
Both, the direct hit	-0.84^{***}	-0.80^{***}	-0.59^{***}
	(0.14)	(0.14)	(0.12)
The rest of the local economy	−0.73	−0.81	0.04
	(0.46)	(0.71)	(0.53)
also without the other manufacturing divisions	−0.36	−0.46	0.14
	(0.42)	(0.71)	(0.55)
All but mining and quarrying (05–09)	-1.22^{**}	-1.38^{*}	−0.39
	(0.46)	(0.70)	(0.54)
All industries	-1.57^{***}	-1.61^{*}	−0.54
	(0.49)	(0.71)	(0.51)
Panel B. Jobs lost for each offshore job lost			
Suppliers	1.14	1.97	2.58
	(0.39)	(0.86)	(2.05)
	[0.5, 2.3]	[0.7, 5.9]	[0.2, 25.9]
The rest of the local economy	1.88	2.99	−0.26
	(1.13)	(2.83)	(3.24)
	[−1.2, 4.9]	[−4.7, 14.4]	[−17.0, 31.1]
also without the other manufacturing divisions	0.92	1.68	−0.88
	(1.08)	(2.75)	(3.24)
	[−1.8, 3.6]	[−6.1, 12.5]	[−19.2, 30.3]
All but mining and quarrying (05–09)	3.13	5.08	2.37
	(1.13)	(3.15)	(3.96)
	[0.4, 6.5]	[−1.3, 18.7]	[−12.0, 52.8]

Note: Panel A as Table 11. The suppliers are the petroleum list's codes other than extraction and support services plus fabricated metal products (25) and machinery and equipment (28), the firms described in Section 5.3, which the sample rule does not remove. The rest of the local economy is all employed residents less the direct hit, and the next row also removes the other manufacturing divisions. Panel B divides each coefficient by the offshore coefficient of the same column, with delta-method standard errors from a stacked regression with region clusters in parentheses and 95 percent percentile intervals from a pairs bootstrap over regions (999 draws) in brackets. Over 2015–19 the offshore loss itself is small and imprecise, so the ratios of column (3) are not informative.

Table 14: Jobs lost for each offshore job lost on the supply-chain samples, 2013–17

	Headline (1)	Supply chain ≤ 2% (2)	Supply chain ≤ 1% (3)
Panel A. Change in employed residents, per 100 employed residents in 2013			
Offshore jobs: extraction and support services (06, 09.1)	-0.39^{***}	-0.31^{***}	-0.35^{***}
	(0.06)	(0.04)	(0.03)
Suppliers: the list's other codes, divisions 25 and 28	-0.45^{***}	-0.39^{**}	-0.47^{*}
	(0.13)	(0.16)	(0.19)
Both, the direct hit	-0.84^{***}	-0.70^{***}	-0.82^{**}
	(0.14)	(0.17)	(0.20)
The rest of the local economy	−0.73	−0.79	−0.15
	(0.46)	(0.90)	(0.85)
also without the other manufacturing divisions	−0.36	−0.62	0.06
	(0.42)	(0.80)	(0.68)
All but mining and quarrying (05–09)	-1.22^{**}	−1.21	−0.63
	(0.46)	(0.87)	(0.86)
All industries	-1.57^{***}	−1.49	−0.97
	(0.49)	(0.91)	(0.89)
Panel B. Jobs lost for each offshore job lost			
Suppliers	1.14	1.26	1.37
	(0.39)	(0.49)	(0.55)
	[0.5, 2.3]	[0.1, 2.5]	[−0.2, 2.9]
The rest of the local economy	1.88	2.52	0.44
	(1.13)	(2.92)	(2.47)
	[−1.2, 4.9]	[−5.9, 7.9]	[−7.9, 6.0]
also without the other manufacturing divisions	0.92	1.98	−0.17
	(1.08)	(2.63)	(1.99)
	[−1.8, 3.6]	[−5.1, 7.2]	[−7.3, 5.0]
All but mining and quarrying (05–09)	3.13	3.89	1.83
	(1.13)	(2.84)	(2.46)
	[0.4, 6.5]	[−4.1, 9.1]	[−6.6, 7.4]
Regions	86	65	55

Note: As Table 13, window 2013–17. Column (1) is the headline sample. Columns (2) and (3) keep the regions with at most 2 and at most 1 percent of their 2003 jobs in SN2002 divisions 11 and 35, extraction and the building of ships and platforms, the supply-chain rule of Section 3.2 and Table 2.

D The 2015 change of source in the employment register

Table 15: The 2015 register break by municipality and offshore work

	Mean	Offshore work, 2003	(SE)	Municipalities
Break, all industries	−3.57	−0.23	(0.33)	375
Break, outside mining and manufacturing	−3.74	−0.24	(0.31)	375
Break, public administration, education and health	−1.52	−0.10	(0.38)	375
Change 2014–15, adjusted	0.75	−0.80***	(0.28)	375
Change 2014–15, unadjusted	−2.83	−1.03***	(0.29)	375
Change 2013–14	1.32	0.06	(0.25)	375
Change 2013–17, bridged across the break	5.00	−1.17	(0.67)	375
Change 2013–17, unadjusted	1.42	−1.40**	(0.59)	375
Change 2015–17	2.93	−0.43	(0.38)	375

Note: Employees aged 15–74 by municipality of workplace in the fourth quarter, mapped to the 2019 municipalities, from the two 2015 counts described in Section 6.5: the unadjusted count on the new source and the count adjusted to be comparable with 2014. The break is 100 times the log difference between them. The changes are 100 times log differences, and the bridged 2013–17 change joins 2013 to the adjusted 2015 and the unadjusted 2015 to 2017. The test is on municipalities, the unit at which the break arises. Mean is the population-weighted mean in the 375 municipalities of the headline sample. The coefficient is on offshore work in 2003 with the controls, weights, inference and stars of column (3) of Table 10.

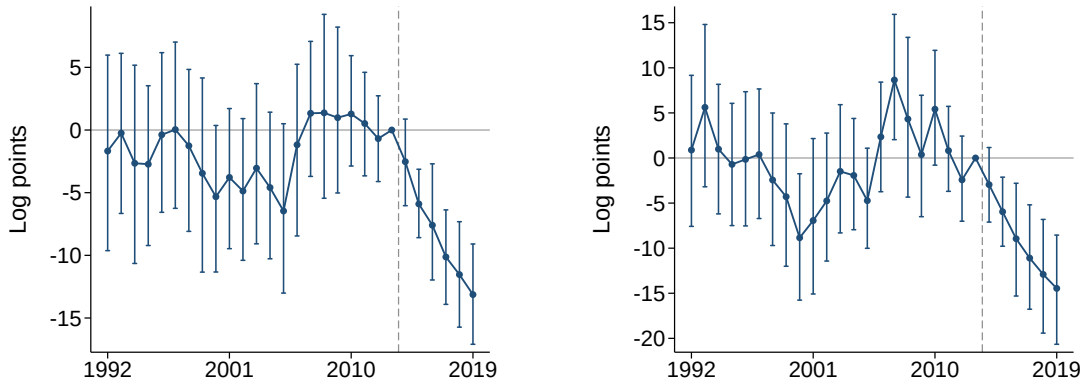
E Pre-boom placebos

Table 16: Pre-boom outcomes on the 2003 offshore share, by sample

	Population growth		Employment	Registered	
Sample	1980–90	1990–2003	growth 2000–03	unemployment 2000–03	Regions
	(1)	(2)	(3)	(4)	
Headline regions	1.91	1.28	0.40	−0.07	86
	(1.12)	(1.18)	(0.47)	(0.14)	
No onshore oil job in the region	1.64	0.09	−0.53	0.78***	58
	(1.87)	(1.94)	(1.01)	(0.25)	
Supply chain 2003 ≤ 2%	2.45	3.29	0.60	−0.05	65
	(1.57)	(1.66)	(0.64)	(0.19)	
Supply chain 2003 ≤ 1%	2.84*	4.21*	1.05*	−0.10	55
	(1.60)	(1.65)	(0.63)	(0.18)	
No hub, 5%	2.81	2.25	0.47	−0.08	66
	(1.39)	(1.41)	(0.48)	(0.15)	
No hub, 3%	1.56	3.78	0.73	−0.01	59
	(1.43)	(1.79)	(0.62)	(0.20)	

Note: Coefficients per percentage point of offshore work in 2003, one observation per region. The controls are those of column (4) of Table 4 with the log population of 2004, and the weights are regional population in 2004. Robust standard errors are in parentheses and stars come from the wild bootstrap. Population growth is 100 times the log change, employment growth 100 times the log change in employed residents from 2000 to 2003, and registered unemployment the change in percentage points, each as the population-weighted mean over the region's sample municipalities. The samples are the 86 regions, the 58 with no onshore oil job in 2003, the regions with at most 2 and 1 percent of their 2003 jobs in SN2002 divisions 11 and 35 (Table 14), and the regions without a hub, a municipality with at least 5 or 3 percent of its jobs on Statistics Norway's petroleum-related list in 2008.

F Further checks



(a) Headline controls **(b)** Coastal controls

Figure 5: The average sale price from 1992, relative to 2013

Note: Coefficients on the region's offshore work in 2003 by year for 100 times the log difference of the average sale price from 2013, with 95 percent wild bootstrap intervals clustered by region, as Figure 3. The sample is the 211 headline municipalities in 81 regions with a price in every year from 1992 to 2019 (Statistics Norway table 06726). Panel (b) adds the coastal controls of column (5) of Table 4. A joint test that the coefficients of 1992 to 2002, with base year 2003, are zero has a p-value of 0.54 in panel (a) and 0.23 in panel (b).

Table 17: The 2003 share and the 2013 share, bust 2013–2017

	Income (1)	Employment (2)	Price (3)
Offshore work, 2003 share	-2.248^{***}	-1.560^{***}	-9.647^{***}
	(0.408)	(0.504)	(1.648)
Offshore work, 2013 share	-1.685^{***}	-0.926^{**}	-6.823^{***}
	(0.344)	(0.422)	(1.547)

Note: Coefficients per percentage point of offshore work with the controls, weights and inference of column (4) of Table 4. The first row is column (4) of Table 4, and the second regresses the same outcomes on the region's 2013 share by least squares. Across regions the 2013 share is 1.22 times the 2003 share and correlates 0.95 with it.

Table 18: The bust with the distance to a large city and centrality, 2013–2017

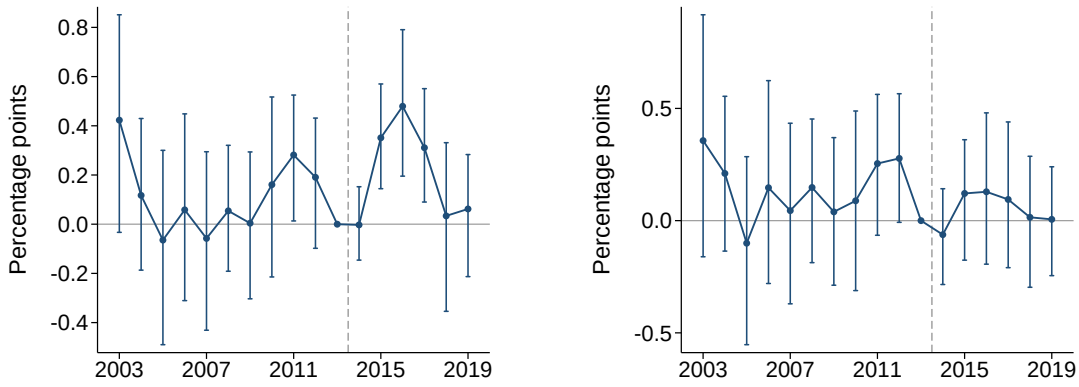
	Income (1)	Employment (2)	Average sale price (3)	Price per m^2 (4)
Headline, column (4) of Table 4	-2.248^{***}	-1.560^{***}	-9.647^{***}	-9.658^{***}
	(0.408)	(0.504)	(1.648)	(1.538)
Distance to a large city	-2.050^{***}	-1.143^{**}	-8.547^{***}	-7.778^{***}
	(0.405)	(0.464)	(1.812)	(1.302)
Distance to a large city and coastal controls	-1.655^{***}	-1.457^{***}	-9.850^{***}	-7.198^{***}
	(0.530)	(0.573)	(2.562)	(1.548)
Centrality index of 2018	-2.154^{***}	-0.965	-8.244^{***}	-8.233^{***}
	(0.435)	(0.544)	(2.055)	(1.654)

Note: Coefficients on offshore work in 2003 with the specification, weights and inference of column (4) of Table 4. The second row adds the regional mean of the log distance to the nearest of Oslo, Bergen, Trondheim and Stavanger, and the third also the coastal controls of column (5). The fourth adds the regional mean of Statistics Norway's 2018 centrality index, in classes from 1 (most central) to 6.

Table 19: The crisis cushion of 2008 and 2009

	Headline (1)	Coastal controls (2)
Employment income per adult man	0.83^{***}	0.65^{***}
	(0.16)	(0.16)
Employment income per adult woman	0.41^{**}	0.34^{**}
	(0.16)	(0.16)
Registered unemployment rate	-0.08	-0.04
	(0.06)	(0.07)
Average sale price	0.36	-1.43
	(2.18)	(2.47)
Detached-house price per m^2	1.91^{***}	0.34
	(0.64)	(1.30)
Number of free-market sales	-0.56	-0.42
	(1.70)	(2.80)

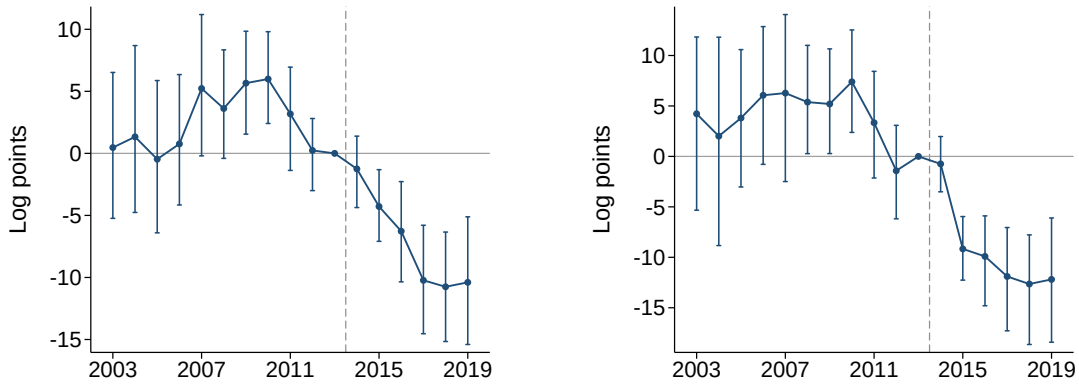
Note: Each entry is $(\beta_{2008} + \beta_{2009})/2 - (\beta_{2006} + \beta_{2007} + \beta_{2010} + \beta_{2011})/4$, the average event-study coefficient of 2008 and 2009 relative to the two years on either side, per percentage point of offshore work in 2003, from region event studies of Equation (2) with base year 2013 over 2005–2013. Incomes and prices are in log points, registered unemployment in percentage points. Column (2) adds the coastal controls of column (5) of Table 4. Standard errors clustered by region in parentheses; stars from the t-test, without the bootstrap. The detached price per square metre uses the municipalities with a price in every year from 2005 to 2013 (64 regions).



(a) Headline controls **(b)** Coastal controls

Figure 6: Registered unemployment, 2003–2019

Note: Coefficients on the region's offshore work in 2003 by year for the change in the registered unemployment rate from November 2013 in percentage points, with 95 percent wild bootstrap intervals clustered by region. Panel (b) adds the coastal controls of column (5) of Table 4. Controls, weights and intervals otherwise as in Figure 3.

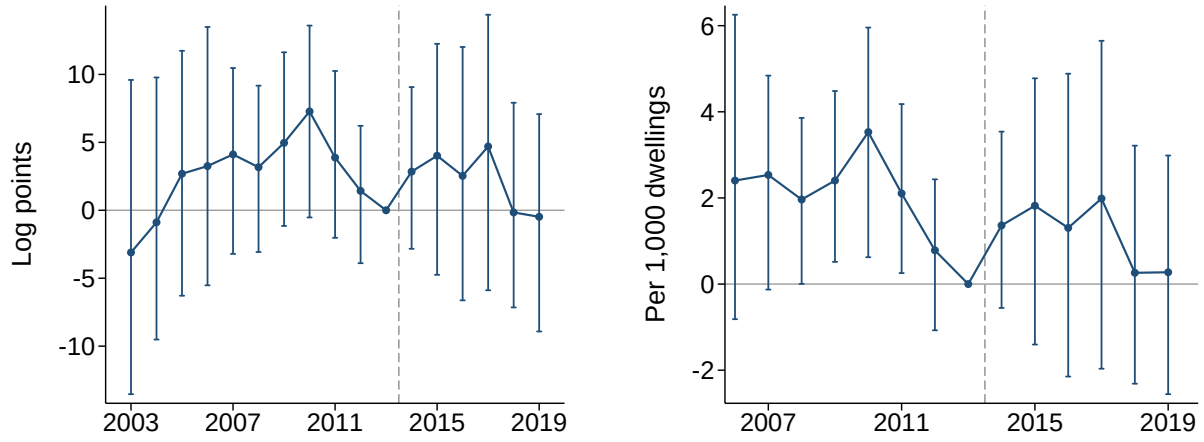


(a) Headline controls **(b)** Coastal controls

Figure 7: The detached price per square metre from 2003, relative to 2013

Note: Coefficients on the region's offshore work in 2003 by year for 100 times the log difference of the detached price per square metre (Statistics Norway table 06035, on the 2019 map) from 2013, with 95 percent wild bootstrap intervals clustered by region, as Figure 3. The sample is the 56 regions whose sample municipalities have a price in every year from 2003 to 2019. Panel (b) adds the coastal controls of column (5) of Table 4. Against the mean of 2003–2006, the coefficient of 2012 is −0.3 (standard error 2.4) in panel (a) and −5.4 (2.9) in panel (b), and that of 2017 is −10.8 (2.7) and −15.9 (3.7).

G Sales volumes



(a) Number of free-market sales, log points

(b) Sales per 1,000 dwellings

Figure 8: Offshore work and the number of dwelling sales, 2003–2019

Note: As Figure 3 in calendar years, for the number of free-market sales and for sales per 1,000 dwellings, both summed over the region's sample municipalities. Sales are 100 times the log difference from 2013. Sales per 1,000 dwellings are the difference from their 2013 value. The second series starts in 2006 with the dwelling stock series.

H Data sources

The supply-chain rule of 2003. Statistics Norway's list of petroleum-related industries (5-digit SN2007, table 13470) starts in 2008. Its correspondence table maps the list into SN2002 as follows: extraction and its support services (06, 09.1) into division 11; the building of ships, platforms and modules, installation work on them and ship repair (30.111, 30.113, 30.115, 30.116, 33.15) into division 35; test drilling (43.13) into 45.120; pipeline transport (49.5) into 60.300; offshore supply shipping (50.204) into 61.106; and supply bases (52.223) into 63.224. Services for the operation of pipelines (52.215, 136 jobs in 2008) have no entry. Of the list's 79,381 jobs located in municipalities in 2008, 90 percent fall into codes that map to divisions 11 and 35. Their share of a region's jobs in 2003 correlates 0.99 with the same share in 2008. Division 35 also contains small boats, leisure boats, rail and aircraft, so the 2003 rule is somewhat broader than the list.

Construction checks. Every analysis dataset is rebuilt from the raw downloads by the R scripts in the project, and a rebuild reproduces them exactly. The checks in `R/11_sanity_checks.R` confirm the following.

- Every national total reaches the 2019 map exactly, except that 22,000 jobs whose place of work is the shelf or unknown have no municipality in the place-of-work register.
- No municipal code with data is lost in the mapping.

- The event-study panel equals the long differences at every point where they measure the same change.

Table 20: Public data sources

Table	Content	Years used
Statistics Norway (StatBank), municipality level		
03311	Employed persons by broad SN2002 industry group (including division 11), by residence and by place of work	2003
03321	Employed persons by municipality of work and of residence (continental shelf)	2000–2019
08545	Employed persons by SN2002 two-digit industry, by place of work (shelf; supply chain, divisions 11 and 35), by residence (fishing, transport equipment, water transport) and by sex (shelf)	2000, 2003, 2008
13470	Employed persons by 5-digit SN2007 industry, place of work (petroleum-related list) and place of residence (industry of workplace, suppliers)	2008, 2013–2019
11365	Employees by place of work and 17 industry groups, 2015 adjusted for the change of source	2013–2015
11396	Employees by place of work and 17 industry groups, unadjusted	2015–2019
06944	Household income after tax and total, median, all households	2005–2019
11607	Employed persons 20–66 by immigrant background	2001–2019
11616	Employed residents 15–74	2000–2003
10540	Registered unemployed 15–74 (NAV), per cent	2000–2019
07221	House price index for existing dwellings, national	2003–2017
13563	Residents 15–74 by labour-market status	2008–2019
06913	Population on 1 January, births, deaths, growth	1970–2019
06265	Dwellings by building type, 1 January	2006–2019
05940	Dwellings started and completed	2000–2019
06035	Price per m^2 and number of sales, by dwelling type	2002–2019
06726	Free-market sales of dwellings, number and value	2000–2019
03068	Average income from employment and gross income per resident aged 17 and over, by sex (tax records)	2005–2019
12314	Employees, jobs and earnings by industry division (88 groups, SIC2007), national	2016
08050	Planned and actual petroleum investment (national)	2002–2019
04170	Accrued petroleum investment, older series (equal to 08050 where both exist)	2000–2019
14701	Consumer price index, annual	2000–2019
Labour and Welfare Administration (NAV)		
Benefits	Unemployment benefits (dagpenger) by municipality: recipients during the year and amounts paid	2015–2018
Statistics Norway, other		
Census	1980 Population and Housing Census, municipal booklets, table 9 (employed persons by place of work; row "North Sea")	1980
Statistics Norway, classifications (KLASS)		
131	Municipality code changes	1970–2019
108	Economic regions (2018 version, 2019 map)	2019
Corr. 108	Correspondence table SN2007–SN2002 (a different series of numbers from the classifications)	2009
Norwegian Offshore Directorate (FactPages and FactMaps)		
Fields	Outlines of the 139 oil and gas fields	current
Other		
Eurostat GISCO	County boundaries (NUTS 2016 level 3)	2016
FRED (EIA)	Brent spot price, daily (DCOILBRENTEU)	2000–2019
Wikidata	Municipal administrative centres	current
Heliports	Coordinates of the five offshore heliports in use in 2003	2003